



Short communication

The effects of introducing withholding and third-party reporting on tax collections: Evidence from the U.S. state personal income tax[☆]Sutirtha Bagchi^a, Libor Dušek^b^a Villanova University, United States^b Charles University, Faculty of Law, Czech Republic

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ABSTRACT

This paper examines the impact of introducing withholding of the personal income tax by state governments in the U.S. We exploit the staggered adoption of withholding by individual states over the period 1948–1987 to construct event study estimates. We obtain a robust finding: Introducing withholding led to an immediate and permanent increase in income tax revenues by about 29 percent, holding tax rates constant. The result is consistent with the crucial role of withholding and third-party reporting in improving tax compliance. We consider several alternative explanations such as changes to the tax base and increases in enforcement activity but these explanations lack support. There is some evidence that non-filing substantially decreased following the introduction of withholding.

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1. Introduction

A growing literature in public economics emphasizes the role of information reporting and withholding in tax collection. Slemrod (2019) notes that the noncompliance rate in the United States varies widely, being as low as 1 percent when there is both information reporting and withholding such as for wages and salaries but as high as 63 percent when there is no third-party information

reporting nor withholding as in the case of self-employment income. Withholding by employers, in particular, plays a critical role in the administration of the tax system. For example, for the United States, withholding of individual income and Federal Insurance Contributions Act (FICA) taxes amounted to \$2.5 trillion or about 70 percent of all federal taxes collected by the IRS in 2019 (IRS Databook 2019, p. 11). As a result, U.S. government officials have described withholding as “the cornerstone of the administration of our individual income tax” and as being “absolutely at the heart of our self-assessment technique of paying taxes” (Twight, 1995).

Instead of collecting income taxes with taxpayers “voluntarily” filing their tax returns, withholding requires employers to deduct the tax from workers’ paychecks and remit it directly to the government. Withholding, as the term is understood, combined two innovations: third-party reporting (the tax collectors being able to verify the employees’ tax records with information from employers) and withholding as such (collecting the tax from the source of the income rather than its recipient). Withholding was introduced for the federal income tax on July 1, 1943 during the course of World War II. The states did not follow suit immediately

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E-mail addresses: sutirtha.bagchi@villanova.edu (S. Bagchi), dusekl@prf.cuni.cz (L. Dušek)

for their state personal income taxes. The adoption of withholding by individual states was staggered and varied widely from 1948 (Oregon) to 1987 (North Dakota), although much of the activity was concentrated in the second half of the 1950s and the early 1960s.

The main research question that we answer in this paper is: What were the causal effects of this important policy reform – the introduction of withholding by U.S. states – on state personal income tax revenues? We exploit the staggered adoption of withholding by individual states that had a personal income tax as of 1944 and introduced withholding subsequently to generate event study estimates of the causal effects. Using a panel dataset of 29 states between 1944 and 2000, we find that withholding led to an increase in real per capita income tax revenues by about 29 percent. This increase in collections occurs precisely at the time of adoption and is permanent. As we control for tax rates (and brackets) in our most complete specification, this estimate has the interpretation of an improvement in the governments' ability to collect the tax. The findings cannot be explained either by changes to the tax base or by varying levels of enforcement activity. We also find no evidence of spillover effects onto other taxes – whether federal or state. Instead our results point to withholding improving compliance with the state personal income tax and therefore driving higher collections. We have suggestive evidence (based on more detailed data from a single state – California) that the increased collections resulted partly through a reduction in non-filing.

Our paper most directly contributes to a recent literature that documents the importance of withholding for tax compliance (e.g., Slemrod and Velayudhan, 2018 and Waseem, 2019). This literature finds increases in tax collections through a default payment channel and a change in enforcement perceptions (Brockmeyer and Hernandez, 2019) as well as through behavioral channels such as reducing frictions in the remittance of taxes (Boning, 2018).¹ We also relate to a larger literature that emphasizes the role of third-party information reporting. Building on earlier work (Gordon and Li, 2005; Kopczuk and Slemrod, 2006), Kleven et al. (2016) provide a theoretical framework to show how increasing firm size makes it difficult to sustain collusive arrangements between the taxpayer and her employer. Empirical evidence consistent with this observation is provided by Besley and Persson (2014) and Jensen (2019) who draw on the historical experience with the personal income tax and argue that there is a causal link between increasing formalization and the ability of governments to collect taxes.²

Our paper has a different focus. It is the first study that analyzes the introduction of withholding for the personal income tax by the U.S. states and derives causal estimates of its *aggregate* impact on tax collections, the government's key variable of interest. The aggregate impact should be interpreted as a joint effect of third-party reporting and withholding itself. The joint effect is of primary policy interest as the implementation of withholding typically encompasses both features. Our second contribution is that while much of the literature examines issues of tax compliance in a developing country context, we examine these issues in a

developed country context but nevertheless find that third-party information reporting and withholding are critical here as well. The findings are economically significant. In fact, it is difficult to think of policy measures that could increase collections from a similarly broad-based and fiscally important tax by the magnitudes on the order of the 29 percent that we find.

The rest of the paper is organized as follows. Section 2 provides institutional background on tax withholding. Section 3 describes the data and our empirical specification. We present our results in Section 4, with possible mechanisms discussed in Section 5. Section 6 concludes.

2. Institutional background³

The introduction of income tax withholding in 1943 by the federal government marked a major change in tax administration. Until then, taxpayers remitted their income taxes when they filed annual returns. Since then, employers have been required to withhold a certain fraction of their workers' paychecks and remit it directly to the government. At the end of the tax year, taxpayers file income tax returns in which they compute their tax liability. Any difference between their tax liability and the amount withheld is either refunded to the taxpayer (if negative) or remitted to the government (if positive).

The introduction of withholding has the potential to increase tax revenue primarily by making tax evasion more difficult (Kleven et al., 2016). The tax authorities receive the tax remittances from firms as the income is earned and do not need to rely on taxpayers' self-reports. Withholding strengthens both negative and positive incentives to file a tax return: a higher probability of detection as well as an incentive to claim a refund as the system is designed to over-withhold on average (Jones, 2012). Additionally, it is more cost effective to collect tax liability "from a small number of organizations with relatively efficient bookkeeping done for nontax reasons" (Slemrod, 2019).

Some limited forms of withholding had existed at the state level before 1943, typically for non-residents. In 1948, Oregon became the first state to require employer withholding on all wage and salary income. Nine states adopted withholding in the following 10 years, five states in 1959, and 12 states during the 1960s (see Table A.1). Except for North Dakota, all 29 states that had a broad-based income tax in 1944 implemented withholding by the early 1970s, with North Dakota eventually introducing it in 1987. These developments occurred as state personal income taxes were gaining importance as a source of revenue. They represented less than 9 percent of tax revenues in the 29 states that were imposing them (and 5 percent in all states) in 1944, the first year of our analysis. Today, by contrast, personal income taxes are the single most important source of revenue for state governments, representing 37 percent of states' total tax revenue.⁴

3. Empirical Analysis

3.1. Construction of Variables and Data Sources

Our dataset is an annual panel of the 29 U.S. states from 1944 till 2000 that had a personal income tax as of 1944 and introduced withholding subsequently.⁵ The data on government tax revenues

¹ Withholding additionally likely affects taxpayer compliance in that individuals who are under-withheld are more likely to under-report or claim deductions. Evidence to that effect is provided by Chang and Schultz (1990) who, using Taxpayer Compliance Measurement Program (TCMP) data, show a correlation between the voluntary compliance rate and whether a balance or refund is due, by Rees-Jones (2017) and Engstrom et al. (2015) who use observational data for individual taxpayers from the U.S. and Sweden respectively, and by Vossler et al. (2021) who substantiate these findings in an experimental setting.

² Another strand of the literature relying on randomized field experiments (e.g., Pomeranz (2015)) or quasi-experiments (e.g., Naritomi (2019)) provides further evidence on the importance of third-party information reporting on tax compliance. Kleven et al. (2011) report the results of a field experiment in Denmark and find negligible underreporting for income subject to third-party reporting but substantial underreporting for income not subject to such reporting.

³ The historical information in this section builds on Penniman (1980).

⁴ Source: Census of Governments (2017). General sales taxes and excise taxes contribute about 32% and 16% of total tax revenue respectively.

⁵ 11 states (e.g., Michigan) are excluded as these states adopted withholding simultaneously with the introduction of the personal income tax, 8 states (e.g., Texas) are excluded as they do not have a broad-based income tax, and Alaska and Hawaii are excluded as fiscal data are only available from 1959 by which point both states had adopted withholding.

(by source) are taken from “Compendium of State Government Finances”, an annual publication of the U.S. Census Bureau. The information on when each state introduced the income tax and implemented withholding comes from Penniman (1980). Data on personal income tax rates and brackets, along with tax base variables (e.g., personal exemption, standard deduction, etc.), are based on Bakija (2017). The Michigan World Tax Database and The Book of the States, a publication of The Council of State Governments,⁶ are the sources of data on rates for the corporate income, sales, and excise tax, with any differences between the two sources reconciled through an extensive search of newspaper archives.

A variety of socio-economic control variables are used in the regressions but sources for these are standard in the literature and are accordingly listed in notes to Table A.2 that provides summary statistics for all variables.

3.2. Empirical Specification

We use an event study design to estimate the impact of withholding. We start off with a parsimonious specification that only includes state and year fixed effects and progressively augment it with socio-economic controls and income tax parameters. Our most complete specification is:

$$Y_{st} = \alpha + \beta_1 * Post_{st} + \beta_2 * X_{st} + \lambda_s + \mu_t + \varepsilon_{st} \quad (1)$$

where: Y_{st} are log of real per capita state personal income tax revenues for state s in year t ; $Post_{st}$ is a dummy variable set to 1 for all years for which withholding is in effect in state s and 0 otherwise; X_{st} includes socio-economic controls, viz. (i) real income per capita, share of personal income derived from (ii) farms and (iii) ownership of capital, (iv) population, (v) share of the population that is elderly, (vi) share Black, and (vii) share urban; τ_{st} are six controls for the key income tax parameters – (i) the top and (ii) bottom income tax rates, (iii) the income levels at which the top rate starts (“top bracket”) and (iv) the bottom rate ends (“bottom bracket”), and the interaction of the (v) top and (vi) bottom tax rates with their respective brackets. λ_s and μ_t are state and year fixed effects, with ε_{st} the error term. Because withholding is introduced at the state level, we cluster standard errors by state to account for intertemporal correlation in the error terms (Bertrand et al., 2004), with robustness checks to the level of clustering presented in Table A.5. While a growing literature has expressed concerns about the validity of the conventional two-way fixed effects estimator, we find those concerns to have less bite in this case as the institutional context suggests the lack of anticipation effects⁷ and our estimates suggest the lack of dynamic treatment effects. These issues are discussed at length in Section C of the Appendix.

An important issue in the context of estimating Eq. (1) using OLS is the possible endogeneity of the introduction of withholding itself. As withholding is meant to improve tax compliance, states may have introduced it when their public finances were under stress and tax collections were temporarily depressed. In that case,

⁶ The publication describes legislative changes occurring in each calendar year, which makes it possible to fill in the tax rates for those years that are not reported in biennial tables.

⁷ Examining the history of these laws shows that enactment of withholding was rarely guaranteed and moreover, the laws were effective within a few months of passage reducing the possibility of anticipation effects. In some cases, the state Supreme Court had to weigh in and opine that the passage of the withholding law was constitutional (as occurred in the case of Oregon) whereas in other cases, enactment of withholding followed from an unusual confluence of events. For instance, in the case of California, Governor Ronald Reagan had gone on the record as having been opposed to withholding but he reluctantly agreed to sign withholding legislation as part of a deal with the Democratic majority on legislative reapportionment (p. 13, The Chico Enterprise-Record, 31 Dec 1971). Moreover, even as withholding was signed into law by Governor Reagan on 8th December 1971, it came into effect in less than four weeks from 1st January, 1972 (p. 3, Times-Advocate, 09 Dec 1971).

tax collections may increase after withholding is adopted but that may simply reflect a reversion to the mean rather than the causal impact of withholding. In order to address this issue, we examine the trends in personal income tax collections around the time of introduction of withholding and examine if the change in collections occurs precisely when withholding is adopted.

A final word about our empirical approach: Our object of interest in this paper is the change in collections of the personal income tax, holding rates and other parameters of the income tax constant. A complete model of the tax policy choices faced by governments would however recognize that because withholding enables the government to collect more revenue at given tax rates from one tax source (the personal income tax), the adoption of withholding is likely to trigger an adjustment in the equilibrium tax rates and the level and composition of tax revenues. That is in fact the focus of our complementary paper (Dušek and Bagchi, 2018) which examines the impact of introducing withholding on the overall state tax system. In that paper we find that following the adoption of withholding, state governments started relying on the personal income tax more for raising revenues compared to other taxes. We also find that while state tax revenues increase significantly, there are no statistically significant increases in state expenditures; this apparent anomaly is explained by the fact that state debt levels stabilize once they adopt withholding. For ease of presentation this present paper maintains a narrower focus and analyzes the causal impact of introducing withholding on the state personal income tax with the subsequent adjustment in the overall tax system the focus of our complementary paper.

4. Results

We first present a graphical overview of the key data on income tax revenues before and after the adoption of withholding in the top panel of Fig. 1. It shows the real per capita personal income taxes, averaged across the 29 states that had an income tax in 1944. All state years on the horizontal axis are normalized so that year zero is the fiscal year in which a state implements withholding. While the figures have to be read with caution as no other factors affecting tax revenues are controlled for, they suggest that withholding had a sizable impact. Real per capita income tax revenues grow slowly but steadily before withholding until they reach 60 dollars, then they discontinuously jump up to 85 dollars immediately after withholding is adopted, and later continue to grow at a similar pace as before.⁸

4.1. Baseline effects of the adoption of withholding on personal income taxes

Table 1 presents our baseline estimates of the causal effects of adopting withholding on the log of real per capita personal income tax revenues. We start off with a parsimonious specification that only includes state and year fixed effects. We add socio-economic controls in col. (2), the top personal income tax rate in col. (3), and top and bottom rates, brackets, and the interaction of rates with the respective brackets in col. (4).

In our most complete specification that includes controls for socio-economic characteristics and all income tax parameters, we obtain a coefficient of 0.290 on the withholding dummy. This coefficient of 0.290 implies that state governments began collecting an

⁸ Because Oregon and Delaware adopted withholding in 1948 and 1949 respectively, we do not have 8 years of data prior to their adoption. Accordingly, the underlying data used in constructing Fig. 1 are unbalanced. For robustness, we provide a version (Figure A.1) that only uses data from states with at least 9 years of data pre- and post-withholding. Additionally, we provide “balanced panel” versions of our estimates in Table A.3. Those estimates guard against changes in group composition driving our results.

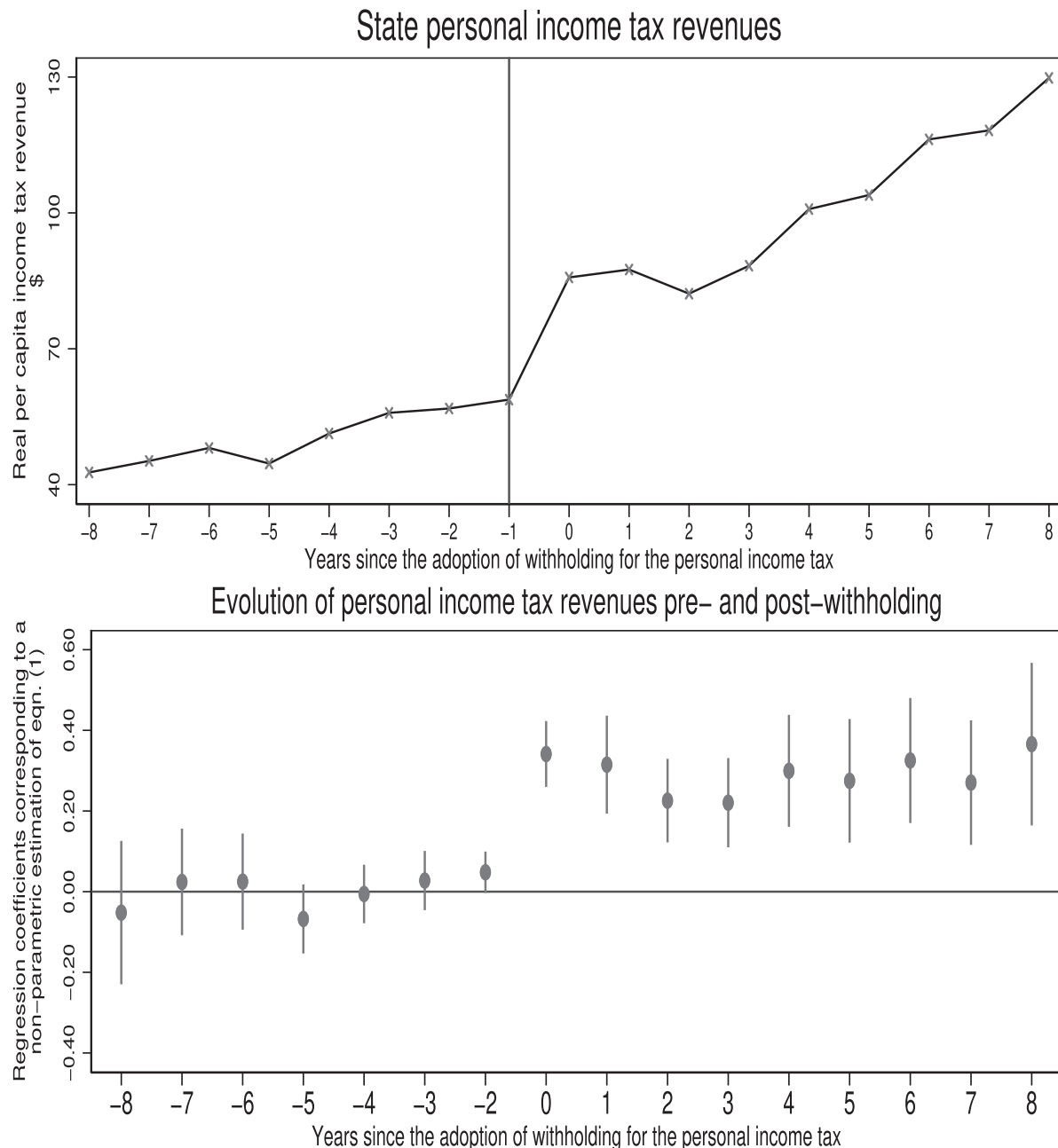


Fig. 1. Trends in state personal income tax revenues around the time of adoption of withholding. Notes: The top panel plots the average real per capita income tax revenue for all 29 states which adopted withholding after introducing the personal income tax. The bottom panel plots the coefficients, along with the 90 percent confidence intervals, from a regression of the log of real per capita income tax revenue on a set of dummy variables, state and year fixed effects, all socio-economic controls, and income tax parameters, with standard errors clustered at the state level. A balanced panel version of this figure is provided in the Appendix as Figure A.1.

additional 29.0 percent in personal income taxes following the adoption of withholding *holding tax rates constant*. The estimates are somewhat larger at about 34.4 percent when we employ a more parsimonious – but defensible – specification in col. (2) which includes state and year fixed effects and all socio-economic controls.

To gauge the dynamics of the response to withholding, we decompose the post-withholding years for each state into three periods in Panel B: a short-run defined as the first 5 years after adoption (including the year of); a medium-run defined as the next 6 years after adoption; and a long-run beyond the first 11 years from the introduction of withholding.⁹ The dynamics of the

response are interesting: in the first two columns, when tax rates are not controlled for, the long-run estimates are the largest in comparison to the short- and medium-run estimates. However, once rates are controlled for, the differences in estimates across the periods shrink, with the short-, medium-, and long-run impacts all roughly equal to each other.¹⁰

We interpret our estimates from Table 1 as the causal impact from the introduction of withholding on tax revenues through improved tax compliance. The results presented in subsequent

⁹ Our results are robust to defining the short-, medium-, and long-run in alternative ways and those robustness checks are provided in Table A.4.

¹⁰ Other than the short-run estimates in col. (1) which are significantly different from the medium-run and the long-run estimates at the 5% level, none of the other pairwise comparisons indicate a statistically significant difference.

Table 1
Baseline estimates of the effects of withholding on the personal income tax revenues.

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Panel A: Baseline specification				
Withholding dummy	0.392*** (0.0809)	0.344*** (0.0809)	0.298*** (0.0799)	0.290*** (0.0763)
Real income per capita		1.001*** (0.225)	1.340*** (0.239)	1.340*** (0.204)
Share of farm income		0.144** (0.0600)	0.130** (0.0538)	0.124** (0.0566)
Share of capital income		0.100 (0.146)	0.105 (0.117)	0.154 (0.120)
Population		0.0803 (0.0914)	0.105 (0.0769)	0.0741 (0.0849)
Percent elderly		−0.383 (0.291)	−0.378* (0.216)	−0.477** (0.201)
Percent Black		−0.266*** (0.0675)	−0.194*** (0.0625)	−0.177** (0.0758)
Percent urban		−0.381 (0.233)	−0.0561 (0.185)	−0.0727 (0.207)
Top personal income tax rate			0.351*** (0.0546)	0.669 (0.508)
Lowest personal income tax rate				−0.275 (0.379)
Top personal income tax bracket				0.0227 (0.109)
Lowest personal income tax bracket				−0.104* (0.0540)
Top personal income tax rate				−0.0330 (0.0547)
X top personal income tax bracket				0.0475 (0.0426)
Lowest personal income tax rate				
X lowest personal income tax bracket				
Panel B: Dynamics of the response to the introduction of withholding				
Short-run effect (between 0–4 years after adoption)	0.362*** (0.0736)	0.326*** (0.0715)	0.289*** (0.0722)	0.283*** (0.0688)
Medium-run effect (between 5–10 years after adoption)	0.474*** (0.113)	0.399*** (0.118)	0.331*** (0.113)	0.319*** (0.106)
Long-run effect (more than 10 years after adoption)	0.582*** (0.133)	0.461*** (0.152)	0.362** (0.143)	0.360*** (0.130)
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓
Observations	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29
Time period included in the estimation	1944–2000	1944–2000	1944–2000	1944–2000

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1% State and year fixed effects are included in all columns but the corresponding coefficients have not been shown. The fiscal and socio-economic control variables included in columns (2)–(4) of both panels A and B are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.

sub-sections are consistent with this interpretation and not with the alternative explanations that we consider.

4.2. Checking for pre-existing trends and additional robustness checks

A fundamental concern in any event study design are possible differences in pre-existing trends. If, for example, an unobservable factor that is positively correlated with the government's ability to collect taxes was operating a few years before a state adopted withholding, the coefficient on the withholding dummy variable could be capturing the effect of such an unobservable factor rather than solely reflecting the causal response to withholding. We address this important concern in two ways.

First, to check for possible unobservable trends and to obtain additional insights about the speed of response of tax revenues to withholding, we estimate the same equation as in Table 1 but replace the withholding dummy with dummy variables representing the year of adoption, the 1st, 2nd, etc. up to 7th year after the adoption of withholding, and a dummy variable equal to 1 for 8 and more years after adoption. We also include dummy variables for the 2nd, 3rd, etc. up to 7 and another dummy variable for 8

(or more) years before the adoption of withholding.¹¹ The coefficients on the dummy variables and their 90% confidence intervals are plotted against the time axis in the lower half of Fig. 1.

Fig. 1 shows that there was no unobservable trend in personal income tax revenues before the adoption of withholding given that the coefficients are fairly precisely estimated and yet not statistically different from zero. The effect of withholding also appears immediately and is persistent: income tax revenues rise by over 30 percent in the adoption year and the effect sizes in the following years are not statistically different from the adoption year. A (statistically insignificant) drop that is visible in the second and third years is due to a particular institutional feature. In most states, taxpayers had to remit the newly withheld taxes as well as the last year's tax liability, generating a one-time fiscal windfall on top of the real improvement in tax collections. This transitory shock is, on average, spread across the adoption year and the first post-adoption year because the date of adoption did not typically coincide with the beginning of the fiscal year. The immediacy of the

¹¹ The dummy variable representing the last year before adoption is dropped so that it serves as the baseline.

effect is noteworthy; one could plausibly expect the effects to build up more gradually as the states learnt to administer withholding and improve compliance. However, to the extent such learning was present, it is empirically dwarfed by the one-time windfall. The sharp jumps make it unlikely that the increased collections were the product of other changes afoot when states adopted withholding; changes in the desired public spending, for example, are likely to have been less abrupt than the break in trend we observe for income tax collections.

In our second approach to address concerns about pre-existing trends, we implement a permutation test in which we take the set of adopting states, and randomly assign to each state a year of implementation.¹² We then estimate our main regression and record the coefficient on the withholding dummy. We repeat this exercise 5,000 times, and investigate the share of placebo coefficients that are larger than the baseline estimates. The results are presented graphically in Figure A.2 and they show that our estimated impact of withholding is much larger than the placebo estimates and underlying trends in tax collection – strong as they were – are unlikely to confound our estimate.

Finally, in Table A.6 we present a battery of robustness checks that further address the issue of pre-existing trends as well as additional concerns. A complete description of these checks is provided in Appendix B but in brief, the checks (i) find no pre-trend in the years immediately preceding the adoption of withholding; and show that our estimates are robust to (ii) omitting transitory shocks in the first two years after adoption, (iii) including state-specific linear or quadratic time trends, (iv–v) including region-by-year or division-by-year fixed effects, (vi) weighting observations by state population, (vii) using a random effects specification, and (viii) distinguishing between states that did and did not forgo part of the tax liabilities in the year when withholding was adopted. Altogether, we believe that our robustness checks provide strong support for our key result that withholding significantly improved tax compliance.

4.3. Alternative explanations for our results

4.3.1. Effect of varying enforcement efforts on compliance

A potential concern with the estimated impact is that states adopted withholding when they were under particular fiscal stress and therefore also enhanced enforcement to reduce non-compliance in which case our estimates are upward biased. While measures of enforcement (such as the number of audits or the size of the enforcement budget) at the state level are unavailable for this period (and in fact time-invariant differences in administrative capabilities across states are already being captured through state FE), we can examine if time-varying increases in enforcement activity are driving our results. We do so by examining the effect of introducing withholding on state taxes *other than the personal income tax*. Statistically significant increases in revenues from these other taxes would suggest that withholding was one of a suite of measures adopted by state governments whereas not finding a significant change in the other revenue sources makes that possibility less likely. Accordingly in Table 2, we examine the effect of the adoption of withholding on three of the most important sources of revenue for states – the corporate income tax, the general sales tax, and the selective sales (excise) tax. The order in which specifications are presented closely mirror those in Table 1.

Examining the coefficients in Table 2, we find that the introduction of withholding did not result in a statistically significant increase in any of these other taxes used by state governments.

¹² The adoption years are drawn from a normal distribution which appears to fit the adoption pattern fairly well.

These results help rule out the possibility that the adoption of withholding coincided with governments simultaneously introducing other measures intended to improve compliance or that there were spillover effects of withholding on these other taxes.

4.3.2. Changes to the tax base

A potential confounding factor that might affect tax compliance is variations in the personal income tax base. If states were broadening their income tax base systematically around the time that they introduced withholding, then our estimates of the effects of withholding could be biased upward. In practice, changes to the tax base are at least as common as changes to statutory rates for the corporate income tax (Suárez-Serrato and Zidar, 2018) and the impact of the Tax Reform Act of 1986 was driven in large part by changes to the tax base (Kopczuk, 2005). Accordingly in Table A.7, we examine the degree to which our estimates are confounded by changes in the personal tax base around the time of adoption of withholding. Among the large number of variables capturing variations in the tax base, we focus on eight of the most salient features of any income tax system including the standard deduction, the personal exemption, and the number of different tax brackets.

We introduce these base variables – either individually or simultaneously in specification (1) – and examine the robustness of the coefficient on the withholding dummy. As shown in Table A.7, the estimated effects are practically unchanged and they confirm that per capita income tax revenues went up by about 28 percent following withholding. Further, we directly examine whether any of these features of the income tax changed discontinuously around the time of adoption of withholding. We however do not find any such impact (Refer Table A.8).

4.3.3. Other confounders – Wage and salaried employment

Over our sample period, U.S. states witnessed a significant growth in tax collections driven by an increase in personal income taxes and they also underwent a transition from self-employment to employee-jobs. Furthermore, as documented carefully by Jensen (2019), the two phenomena are causally linked; wage and salaried employment results in the generation of information trails that then make it possible for governments to audit taxpayer reports of income. Consequently, a change in the structure of the economy towards more employee-employment followed by reductions in the filing threshold could confound our estimates.¹³

We address this concern in two ways. We examine the impact of introducing withholding on the proportion of state personal income derived from wages and salaries and the percentage of workers who work for wages and salaries (as opposed to self-employment or unpaid work). Those estimates are economically small and indistinguishable from zero. Also, we conduct a sample split based on a state's participation in the Industrial Development Bonds (IDB) program, implemented across states in 1950s–1960s and exploited as an IV in Jensen (2019).¹⁴ Our results in Table A.9 indicate that the impact of withholding was identical across the 22 states that introduced this program and the 7 states that never introduced it. Furthermore, in the 22 states that introduced the IDB program, there is no differential impact of withholding following the decision by the respective state supreme court to uphold the IDB program.

¹³ The robustness of our findings to the inclusion of state-specific trends in Table A.6 partly addresses this concern as changes in the structure of the economy are likely to be gradual and unlikely to exhibit discontinuities at the time when withholding gets introduced.

¹⁴ The IDB program sought to expand industry employee-jobs to areas characterized by under-employment among self-employed, thereby acting as a positive level-shifter in employee-share.

Table 2

Estimates of the effects of withholding on other taxes – testing for the presence of spillovers.

	(1)	(2)	(3)
Panel A: Dependent variable: Log of real per capita corporate income tax revenue			
Withholding dummy	0.150 (0.111)	0.117 (0.101)	−0.0186 (0.0604)
Corporate income tax rate			0.858*** (0.0976)
Observations	1639	1639	1639
Number of states included in the estimation	29	29	29
Time period included in the estimation	1944–2000	1944–2000	1944–2000
Panel B: Log of real per capita general sales tax revenue			
Withholding dummy	−0.0245 (0.0415)	−0.0168 (0.0413)	−0.0195 (0.0352)
General sales tax rate			0.732*** (0.0826)
Observations	1257	1257	1257
Number of states included in the estimation	26	26	26
Time period included in the estimation	1946–2000	1946–2000	1946–2000
Panel C: Log of real per capita selective sales tax revenue			
Withholding dummy	0.0229 (0.0638)	−0.00577 (0.0611)	−0.0143 (0.0539)
Motor fuel tax rate			0.413*** (0.0738)
Observations	1566	1566	1566
Number of states included in the estimation	29	29	29
Time period included in the estimation	1947–2000	1947–2000	1947–2000
State and year fixed effects	✓	✓	✓
Socio-economic controls		✓	✓

Std. errors clustered by state in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. State and year fixed effects are included in all columns but the corresponding coefficients have not been shown. The fiscal and socio-economic control variables included in columns (2) and (3) are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs. The regressions are estimated using the same sample as in Table 1, i.e. the 29 states which had a broad-based personal income tax as of 1944, but of those states, not all employ these other sources of revenue or may not have had the corresponding tax in place for the entire period of the sample. For example, although Delaware is included in the sample in Panel A (corporate income tax), the state adopted a corporate income tax in 1957 and it came into effect in 1958. Likewise, although Georgia is included in our sample in Panel B (general sales tax), it did not adopt a sales tax until 1951. The number of observations included in the estimation in Panels B and C is further limited by the availability of the rates of sales tax (Panel B) and the excise tax on gasoline (Panel C).

5. Mechanisms

Although the large increase in state personal income taxes following the adoption of withholding is unmistakable and holds up to an extensive battery of robustness checks, commenting definitively on the underlying mechanisms that contribute to this large increase is complicated. In the following sub-sections we however offer some tentative explanations.

5.1. Suggestive evidence on the role of withholding itself to the success of this reform

Unlike Brockmeyer and Hernandez (2019) and Waseem (2019) who use rich administrative data from Costa Rica and Pakistan respectively to comment on the differential impact of withholding versus the impact of third-party information reporting, we are unable to do so here. Once a state introduces withholding, both withholding and third-party information reporting come into effect *at the same time*. We can however assess the relative importance of these two channels by exploiting the signing of Audit

Exchange Information Agreements (AEIA), which were agreements between the states and the federal government to exchange information about income tax audit plans and techniques. Under these agreements, states receive revenue agent reports from the IRS that summarize the conclusions from auditing both individual and corporate tax returns (Penniman, 1980). Recent work (Troiano, 2018) suggests that these agreements increased state income tax revenues.

As Table A.1 indicates, there is considerable heterogeneity in when states entered into these AEIAs. As a fortunate coincidence, 15 of the 29 states that we include in our analysis adopted withholding *before* they signed these agreements with the federal government, whereas the remaining 14 states adopted withholding *after* (or simultaneously with) the signing of these agreements. Thus, if the availability of third-party information were critical to the success of this policy reform, we might expect to see a more modest impact of withholding in the latter states where an Agreement with the federal government was already in place.¹⁵ Panels A and B of Table 3 present results from such a sample split but they indicate that the effects of withholding were in fact equal in both sub-samples.

Moreover, when we re-estimate our regressions on the full sample of 29 states and also include the interaction between the withholding and AEIA dummies, we see that although signing an Information Agreement (in the absence of withholding) was associated with a 15.2 percent increase in income tax revenues, the estimated impact of withholding (in the absence of the Information Agreement) remains large (25.6 percent). The negative (albeit statistically insignificant) interaction term suggests that the marginal effect of withholding may be smaller once the AEIA is in place; and vice versa for the effect of the AEIA. However, even if taken at face value, the interaction term implies that the marginal effect of withholding is still 23.0 percent when the AEIA is in place.¹⁶ Additional sample-splits based on whether states adopted withholding early or late in our sample period or whether they had more or less experience with the personal income tax¹⁷ do not indicate statistically significant differences across the various sub-samples. Those regressions included in Table A.10, along with the analysis presented here, suggest that withholding itself played a central role in the success of this policy reform.¹⁸

5.2. The Case Study of California

Being able to dissect whether the increased compliance results from a reduction in non-filing versus increased compliance for filers is challenging as it requires a state-level panel dataset on taxpayers – something that simply does not exist for our sample period. We are however able to offer some tentative evidence by examining the change in the number of returns for one state – California. California was the second last state in our sample to introduce withholding in 1971, making it feasible for us to obtain data

¹⁵ This test admittedly has its limitations. Although the signing of an AEIA enables a state to receive information about those tax returns that were audited, in practice only a small fraction of returns are audited and therefore the state only receives information on those tax returns. In contrast, the introduction of withholding enables a state to receive information on all wage and salaried taxpayers – a population that is likely much larger than just those individuals who had their returns audited by the federal government.

¹⁶ I.e. 0.345–0.115, see Panel C, Column (4) in Table 3.

¹⁷ Quantified by the number of years elapsed between introduction of the PIT and the adoption of withholding.

¹⁸ Also regardless of whether states adopted withholding, employers – particularly large ones – would have had a record of what they were paying their employees and in principle, that information could be requested by state revenue agencies in case of prosecution for tax evasion or fraud. Thus, unlike self-employment income, wage and salary income generates information trails, even when a state is yet to adopt withholding.

Table 3

Estimates of the effects of withholding on the personal income tax, controlling for the adoption of federal-state information sharing agreements.

	(1)	(2)	(3)	(4)
Panel A: States that adopted withholding before information sharing agreements				
Dependent variable: Log of real per capita personal income tax revenues				
Withholding dummy	0.467*** (0.0902)	0.433*** (0.0699)	0.400*** (0.0550)	0.383*** (0.0529)
Dummy for presence of Audit Exchange Information Agreements	0.197* (0.102)	0.159 (0.112)	0.171* (0.0955)	0.169* (0.0862)
Observations	855	855	855	855
Number of states included in the estimation	15	15	15	15
Panel B: States that adopted withholding after (or concurrently with) information sharing agreements				
Dependent variable: Log of real per capita personal income tax revenues				
Withholding dummy	0.306*** (0.0664)	0.262** (0.0912)	0.229** (0.0905)	0.233** (0.0848)
Dummy for presence of Audit Exchange Information Agreements	0.232** (0.0852)	0.200** (0.0891)	0.172* (0.0838)	0.144* (0.0719)
Observations	798	798	798	798
Number of states included in the estimation	14	14	14	14
p-value from a t-test that the effects of withholding are equal across the sub-samples in Panels A and B:	0.1295	0.1126	0.0849	0.1086
p-value from a t-test that the effects of the information sharing agreements are equal across the sub-samples in Panels A and B:	0.7825	0.7575	0.9949	0.8168
Panel C: Dependent variable: Log of real per capita personal income tax revenues, estimated on the overall sample				
Withholding dummy	0.453*** (0.0956)	0.411*** (0.0879)	0.362*** (0.0804)	0.345*** (0.0706)
Dummy for presence of Audit Exchange Information Agreements	0.224*** (0.0638)	0.205*** (0.0661)	0.180** (0.0679)	0.168** (0.0620)
Withholding dummy X Dummy for presence of Audit Exchange Information Agreements	−0.124 (0.0853)	−0.138 (0.108)	−0.129 (0.108)	−0.115 (0.0929)
Observations	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29
Time period included in the estimation in Panels A-C	1944–2000	1944–2000	1944–2000	1944–2000
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%.

The fiscal and socio-economic variables included in cols. (2)–(4) are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.

on the number of income tax returns filed around the period when withholding was adopted. To present the data visually, in the top panel of Fig. 2, we plot the year-over-year change in the number of personal income tax returns filed in the state for the +/- 8-year window around the year of adoption. As a control, we also plot the year-over-year change in the number of *federal* income tax returns filed from California during this period.

Fig. 2 suggests that there was a large and abrupt increase in the number of state income tax returns filed in California in the year in which withholding gets introduced. In that year, the number of state income tax returns jumps from about 5.7 million to about 7.0 million – a remarkable increase of 23 percent – whereas in the 8 years prior to adoption, the number of returns had grown annually at an average of 2.5 percent.^{19,20} Higher filing rates are consistent with withholding's strengthening of the negative and positive incentives to file a return (as discussed in Section 2). In contrast to the sharp increase in the number of state income tax returns in the year of adoption of withholding, Fig. 2 indicates no similar jump

¹⁹ While the largest percentage increase in the number of returns occurs in the year of adoption, the number of returns grows at a faster pace during the remainder of our sample period. This speaks to the immediacy of the impact from introducing withholding, as well as to a possible presence of learning effects (discussed in Section 4.2) that were, however, relatively limited.

²⁰ Although this fact suggests surprisingly high levels of non-filing in the pre-withholding period, Erard and Ho (2001) estimate a non-filing rate of 7 percent with the federal income tax in 1988, even though the IRS has had payroll withholding since 1944 and is likely far better positioned than the average state to enforce tax compliance. Likewise, Meiselman (2018) estimates that 48 percent of Detroit's individual income tax returns in 2014 were not filed on time, even though firms located within Detroit are required to withhold and remit income taxes to the city.

in the number of federal income tax returns filed for California. These results, along with those presented in Table A.11, offer further support for our claim that the adoption of withholding did not result in spillover effects on other taxes – federal or state – but simply reduced evasion of the *state* personal income tax.²¹

The lower half of Fig. 2 provides evidence consistent with substantial non-filing in the pre-withholding period exploiting the limited county-level data available for California for 1968–1973. It indicates that although there was a positive (but statistically insignificant) relationship between the change in the number of state income tax returns filed annually from each county and the adjusted gross income (AGI) as reported on those returns prior to 1972, that relationship turns strongly negative in the post-withholding period. Thus, counties with lower AGIs experienced larger increases in the number of returns filed on average during the post-withholding period.²² This observation is consistent with findings in Erard and Ho (2001) who report that nonfilers have considerably lower incomes than filers and Holcombe and Gmeiner (2020) who find larger overall increases in the number of federal income tax returns filed after the adoption of federal withholding in 1943 lower down in the income distribution.

Overall the results in this sub-section suggest that one channel through which withholding increased tax collections was a reduction in the number of non-filers about whom the government

²¹ Table (A.11) examines the impact of withholding on (1) federal personal income tax and (2) payroll tax collections, and (3) number of federal income tax returns filed from each state and indicate a null result.

²² Corresponding regression results are presented in Table A.12.

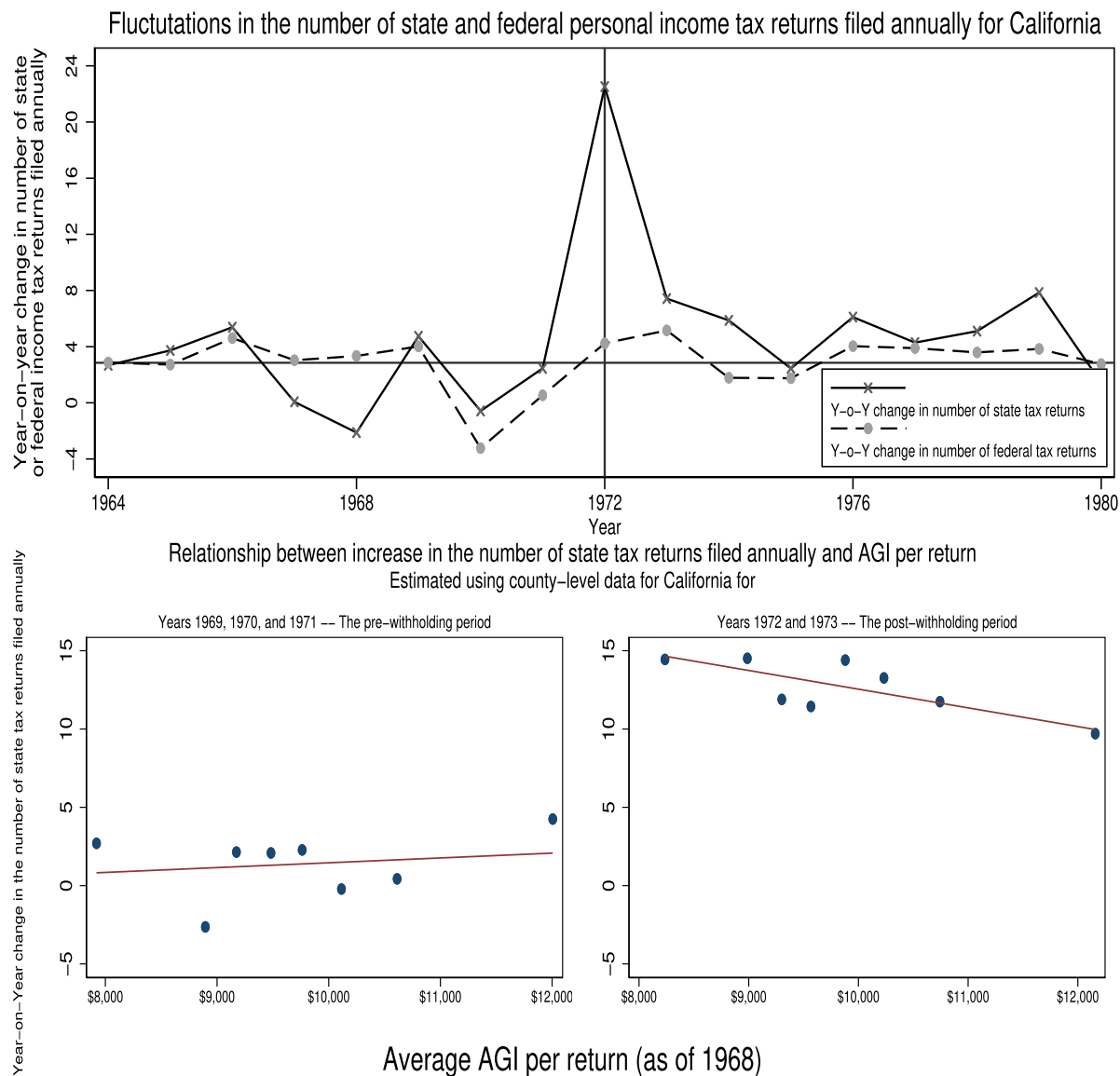


Fig. 2. (a) Annual fluctuations in the number of income tax returns filed for California around the time of adoption of withholding and (b) its relationship with the AGI as of 1968. Notes: The top panel plots the year-on-year change in the number of state and federal personal income tax returns filed for the state of California and illustrates the variation around the year of adoption of withholding in the state (1971). The bottom panel shows scatter plots of the year-on-year change in the number of state personal income tax returns filed annually against the average AGI per return (measured as of 1968), with the figure on the left using data for three years from the pre-withholding period (1969–1971) and the figure on the right using data for two years from the post-withholding period (1972–1973).

would have previously known little and had no tax liability deducted at source.

6. Conclusions

Withholding and third-party reporting have become important themes in public economics. While the existing research clearly demonstrates their key role in tax compliance at the level of individual taxpayers, less is known about their aggregate effects on overall collections. The staggered adoption of withholding at the state level in the U.S. provides a unique setting to fill this gap in the literature. Using an event study approach, we find that withholding led to an immediate and permanent increase in income tax collections by about 29 percent, holding tax rates constant. This is a key and robust finding that survives numerous robustness

checks and alternative empirical strategies. The increase in collections by 29 percent is quantitatively very large; in fact, it is difficult to think of policy measures that could increase collections from a similarly broad-based tax by this much.

We do however caveat our findings in two ways. First, the effects we estimate result from the simultaneous introduction of third-party reporting and withholding. Thus, our results do not directly speak to the estimated increase in revenues that are likely to result from only introducing third-party reporting. Second, we would note that the magnitude of these effects could be smaller in other countries with enforcement capabilities that are less sophisticated compared to that of U.S. states. For instance, although just 1.4% of taxpayers under-reported wage income in Denmark in 2006 (Kleven et al. (2011)), the fraction of wage under-reporters in developing countries are substantially higher, on the orders of 19.3% in Pakistan (Best (2014)) or 18.5% in Uru-

guay (Bergolo et al. (2020)).²³ Thus, to the extent that other governments are less efficient than the U.S. states in tax administration and enforcement, the effects of introducing withholding could be smaller in those other contexts.

These limitations of our study notwithstanding, our findings are of much relevance to policy makers today. For instance, the growth in the “gig economy” generates new sources of income that, under current rules, are largely neither withheld nor reported to tax authorities. For example, more than 2.5 million Americans earned income via on-demand platforms like Airbnb, Etsy, and Lyft in 2014 and the companies generated an estimated \$15 billion in revenues.²⁴ Should tax authorities extend withholding and third-party reporting on such platforms, our paper suggests that the resulting improvements in tax collections could be large.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jpubeco.2021.104537>.

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²³ Both studies quantify under-reporting by comparing wages reported by the taxpayer on her tax return with those reported by their employer. The numbers cited above classify employees as under-reporters if they sub-declare wages by 0.25% or more.

²⁴ <https://www.bloomberg.com/news/articles/2016-05-23/airbnb-others-pay-out-billions-beneath-irs-s-radar-study-finds>

The Effects of Introducing Withholding and Third-Party Reporting on
Tax Collections: Evidence from the U.S. State Personal Income Tax
(Online Appendix)

Sutirtha Bagchi*

Libor Dušek†

October 29, 2021

*sutirtha.bagchi@villanova.edu; Villanova University.

†dusekl@prf.cuni.cz; Charles University, Faculty of Law.

A Empirical approach

This section derives our reduced form regression equation that estimates the causal effects of withholding on personal income tax revenues starting from a simple model. Government collects the personal income tax R^I by imposing a tax rate τ on income y . Because the government is an imperfect tax collector, the revenue actually collected is a fraction $a \in (0, 1)$ of the revenue that would be collected under perfect compliance:

$$R^I = a\tau y \quad (1)$$

The parameter a depends, for example, on the state's economic characteristics and the state's expenditures on tax enforcement. Withholding enters the model as an upward shift in a . Due to the adoption of withholding, we should observe an increase in income tax revenues holding tax rates and income constant.

Next, we derive an econometric model of the personal income tax revenues that reflects the structure of equation (1). For expositional clarity, it abstracts from the complexities of actual income tax schedules such as graduated rates or exemptions and assumes that all income y_{st} is taxed at a flat tax rate τ_{st} . (The estimation issues raised by relaxing this assumption are addressed at the end of this section.). The subscripts s and t denote state and year respectively. All financial variables (revenues and income) are in real per capita terms.

The revenue equation is defined as follows:

$$R_{st}^I = (1 + \alpha D_{st}) r^I(X_{st}, y_{st}) \tau_{st} y_{st} e^{u_{st}^I} \quad (2)$$

The term $(1 + \alpha D_{st}) r^I(X_{st}, y_{st})$ is an empirical counterpart of a in equation (1). D_{st} is a dummy variable indicating whether a state has withholding. The coefficient on the withholding dummy, α , measures the percentage increase in tax collections holding the tax rate constant. The term $r^I(X_{st}, y_{st})$ denotes the fraction of income that the government is able to tax in the absence of withholding. It is a function of socio-economic variables, X_{st} , and the income per capita, y_{st} . u_{st}^I is the error term.

Taking logs on both sides of the equation above, we obtain:

$$\log R_{st}^I = \log(1 + \alpha)D_{st} + \log r^I + \log \tau_{st} + \log y_{st} + u_{st}^I \quad (3)$$

We assume a log-linear functional form for a :

$$\log r^I = a_X \log X + a_y \log y \quad (4)$$

Substituting (4) in (3), we obtain:

$$\log R_{st}^I = \log(1 + \alpha)D_{st} + \log \tau_{st} + a_X \log X_{st} + (1 + a_y) \log y_{st} + u_{st}^I \quad (5)$$

Our empirical specification (1) corresponds directly to the equation above. In our richer specifications, equation (1) is augmented by additional tax code variables to control for complexities of the state income tax systems: (i) the top and (ii) bottom income tax rates, (iii) the income levels at which the top rate starts (“top bracket”) and (iv) the bottom rate ends (“bottom bracket”), and the interaction of the (v) top and (vi) bottom tax rates with their respective bracket (Table 1, Column (4)). We further control, either individually or simultaneously, for numerous tax code parameters affecting the tax base (Refer Table A.7).

B Additional robustness checks

In this section, we check robustness of the finding that income tax revenues increased following the adoption of withholding using a number of tests, in addition to those described in Section 4.2. To varying degrees they address the possibility that there are omitted variables in our analysis which induce an upward bias on the coefficient of interest – the dummy variable corresponding to the adoption of withholding. In the interest of space, we only present the coefficient on this dummy variable in Table A.6, with complete results available on request.

RC1: Pre-trend estimates Beyond the two approaches outlined in Section 4.2 to address the issue of pre-existing trends, we illustrate a third approach here. We examine the trend

in revenues from the personal income tax prior to the adoption of withholding by including in our regressions only observations before the adoption of withholding and a dummy variable for all observations either one or two years before the adoption of withholding, with all other pre-withholding years being absorbed into the constant. The results, presented as Robustness Check 1, reveal no significant pre-trend as the coefficients on the dummy variable corresponding to observations one or two years before the adoption of withholding are statistically insignificant. Thus tax revenues in the two years just before the adoption of withholding are indistinguishable from tax revenues in prior years, which suggests no prior trend in revenues for the personal income tax.

RC2: Ignoring transitory shocks in the first two years after adoption As described in Section 4.2, states experienced a windfall following the adoption of withholding because of a one-time forward shift in the remittance of taxes. This effect appears in the first and second year post adoption because in many states withholding came into effect on January 1, while for most states the fiscal year begins on July 1. Thus the forward shift in tax collections that occurs in the first calendar year after adoption is accounted for in two fiscal years and to reassure the reader of the robustness of our results, we present this check in which we exclude the first two years following the introduction of withholding.

RC3: Including state-specific time trends To account for changes in the structure of the economy such as the share of employees in the labor force as well as improvements in administrative capability, favorable economic trends, increases in the political benefits of public spending, and decreases in the political cost of taxation, we include state-specific linear time trends as a robustness check. Our results are robust to the inclusion of these trends with an estimated impact of withholding of 22 percent in the most complete specification that includes all socio-economic controls and parameters of the income tax system. To the degree that the underlying growth in state taxes is convex, the inclusion of state-specific linear trends could lead to overestimating the ‘post-withholding’ dummy. To address this concern, we implement state-specific quadratic time trends but in fact, we find even larger estimates compared to those with state-specific linear time trends. In either case though, these checks confirm that the introduction of withholding resulted in a large and substantial increase in personal income tax revenues.

RC4: Including region-by-year fixed effects States that adopted withholding were spread across all four Census regions of the country. In a robustness check that is particularly demanding of the data, we introduce region-by-year fixed effects. Our identification thus comes from comparing means for a given state before and after the adoption of withholding allowing for each region to be hit by a unique revenue shock in each year. We find our results for the personal income tax robust to the inclusion of these region-by-year fixed effects.

RC5: Including division-by-year fixed effects In another check that is even more demanding, we introduce Census division-by-year fixed effects for each of the nine divisions and estimate the impact of withholding. Even though some states drop out of the estimation (because there was only one state in that division which had a personal income tax in 1944),¹ the estimated impact of withholding is similar to those in our baseline specification.

RC6: Weighted regressions where the state population serves as weights Our regressions thus far have been unweighted under the premise that the adoption of withholding in each state can be viewed as a policy change of equal importance, regardless of the size of populations in those states. In this robustness check we weight observations by state population but find that our results are practically unchanged. To address any potential concerns about in- and out-migration because of a state's fiscal policies, including, but not limited to, its adoption of withholding, as an additional check, we use weights based on the population at the start of our sample period in 1944 but making that change does not affect our results.

RC7: Use of a random effects specification Our results thus far have included state fixed effects to partial out time-invariant differences that are likely to influence tax compliance. That choice is justifiable on the basis of a Hausman test which rejects the null hypothesis of no correlation between unobserved state-specific random effects and the explanatory variables. However, here we estimate a random effects specification as Griliches and Hausman (1986) stress that observing similar estimates across alternative estimation techniques signals the absence of serious errors-in-variables problems. Using the random effects specification, we find that the adoption of withholding had a similar-sized impact of 30 percent.

¹For example, in the Mid-Atlantic region that includes the states of New York, New Jersey, and Pennsylvania, only New York had a personal income tax at the start of our sample period in 1944.

RC8: Distinguishing between states that forgave part of the income tax due The initial proposals of introducing withholding faced stiff resistance because some taxpayers would in effect have to pay their taxes twice within the span of a year in the form of lump-sum payment for the previous year’s taxes *and* withholding for the current year’s taxes. State governments, in general, did not adopt plans for forgiveness of the income tax liability in the year they adopted withholding, with California, Minnesota, New York, and Wisconsin being the few exceptions (Murray, 1964). In this robustness check, we split states based on whether they forgave a portion of their tax liability or not and largely find similar estimates of the effects of withholding in both groups of states.

The results in Table A.6 offer some confidence in our findings that the adoption of withholding resulted in a large and meaningful increase in personal income tax revenues and omitted variables are unlikely to be driving our results. Moreover, in identical robustness checks for the three other sources of tax revenue examined previously – the corporate income tax, the general sales tax, and the selective sales tax – we confirm that the adoption of withholding did *not* result in an increase for these other taxes validating our view that the effects of withholding on personal income tax revenue we estimate are causal in nature.

C Addressing concerns about two-way Fixed Effects Estimators

A large and growing literature has pointed out the potential pitfalls of using two-way fixed effects regressions (e.g., Athey and Imbens (2021), Borusyak, Jaravel, and Spiess (2021), Callaway and Sant’Anna (2020), Goodman-Bacon (2021), Gardner (2021), and Sun and Abraham (2020)). Our reading of this literature give us reason to pause and reflect on whether the underlying assumptions justifying the use of the two-way FE estimator (TWFE) hold in our context but in the end, we believe that the TWFE estimator does provide us an estimate of the causal parameter of interest, viz. the increase in real per capita income tax revenues following the adoption of withholding. We make that claim for a number of reasons anchoring our discussion with reference to the decomposition in Goodman-Bacon (2021):

$$\text{plim} \hat{\beta}^{DD} = \beta^{DD} = VWATT + VWCT - \Delta ATT \text{ (ref: equation (15), p. 7).}$$

In this expression, $VWATT$ is the “variance-weighted average treatment effect on the treated”, $VWCT$ are “variance-weighted common trends” which generalizes common trends to a setting with timing variation, and ΔATT equals a weighted sum of the change in treatment effects within each timing group’s before and after a later treatment time. For the reasons that we argue below, the last two terms in the above equation can be ignored and the TWFE estimator does recover an average treatment effect on the treated, albeit one that is variance-weighted with weights depending on the “absolute size of the subsample” (number of units undergoing treatment) and “variance of the fixed-effects-adjusted treatment variable in the subsample”. To the extent that all the ATTs are the same, variance-weighting does not matter.

First, as the figures provided in the lower panels of Fig. 1 and A.1 suggest, there was no unobservable trend in personal income tax revenues before the adoption of withholding given that the leads for the pre-withholding period are statistically indistinguishable from zero. The results provided as RC1 of Table A.6 also confirm the lack of any significant pre-trend as tax revenues in the two years just before the adoption of withholding are indistinguishable from tax revenues in prior years. Thus, we contend that the common trends assumption are met here that would result in the second term of the expression, $VWCT$, equaling zero.

We also argue that the third term in the decomposition, ΔATT , can be ignored in our context. As Goodman-Bacon (2021) notes “time-varying effects bias estimates away from $VWATT$ because $\Delta ATT \neq 0$.” Here however, we find little evidence of time-varying effects and we refer back to our results in Panel B of Table 1. Although our decomposition of the post-withholding years into three periods was motivated by the desire to better understand the dynamics of the response, the fact that there are no statistically significant differences between the short-, medium-, and long-run estimates in columns (2) through (4) suggest that the effects of withholding do *not* grow larger with time. Additionally, when we conduct a formal examination of the growth rate in per capita income pre- and post-withholding, we discern no change in growth rates following the adoption of withholding. Thus, withholding causes a discrete jump but not in a change in the growth rate of revenues. Lastly, the sample-split between early and late adopters of withholding which indicated no statistically significant differences across the sub-samples of states depending on when they adopted withholding also suggest that the treatment outcome is *not* influenced by the duration of treatment. Those pieces of evidence, taken as a whole, offer some confidence that the term, ΔATT , stemming

from heterogeneity in treatment effects over time, can be ignored in our context.

Although we have evidence to suggest that the two terms, $VWCT$ and ΔATT , can be assumed to be zero in our context, we undertake the Bacon decomposition as well to better understand where the identifying variation that is exploited by the TWFE estimator comes from.² The Bacon decomposition (Fig. A.3) is reassuring in that it shows that the estimated impact resulting from using earlier adopters as the treated group (and late adopters as the control) is similar to that resulting from using late adopters as the treated group (and early adopters as the control). The two distributions appear similar with random variation around their respective means.

Furthermore, we also subject our results to three robustness checks provided in Goodman-Bacon (2021) which re-examines prior work regarding the impact of unilateral divorce laws. Although Stevenson and Wolfers (2006) find that the passage of unilateral divorce laws leads to a drop in female suicide rates by about 9.7 percent, Goodman-Bacon (2021) finds those results not robust to either the inclusion of state-specific linear time trends or region-by-year fixed effects or weighting states by population (Refer Table 2, p. 15). In contrast, as the robustness checks provided in Table A.6 indicate, our results survive the introduction of state-specific linear time trends (RC3.1), region-by-year fixed effects (RC4), and the use of weighted least squares (RC6.1 and RC6.2).

Although we believe – and have attempted to justify that belief – that the TWFE estimator is valid in our context, we also confirm our results through the use of a newly developed estimator outlined in de Chaisemartin and D’Haultfoeuille (2020) that is robust to heterogeneous or dynamic treatment effects. Using the “did_multiplegt” estimator, we estimate an immediate impact of about 35 percent in tax collections following the adoption of withholding with an overall average effect of about 22 percent, regardless of whether we include controls or not (Ref. Table A.13). Moreover, when we include leads in our empirical specification to test for parallel trends (or the lack thereof), we again confirm our OLS results, viz. the leads are indistinguishable from zero. Lastly, we generate an event study plot using this package and include it as Figure A.4. The figure confirms the lack of a pre-trend prior to the adoption of withholding and a sharp rise in collections right when withholding is adopted, consistent

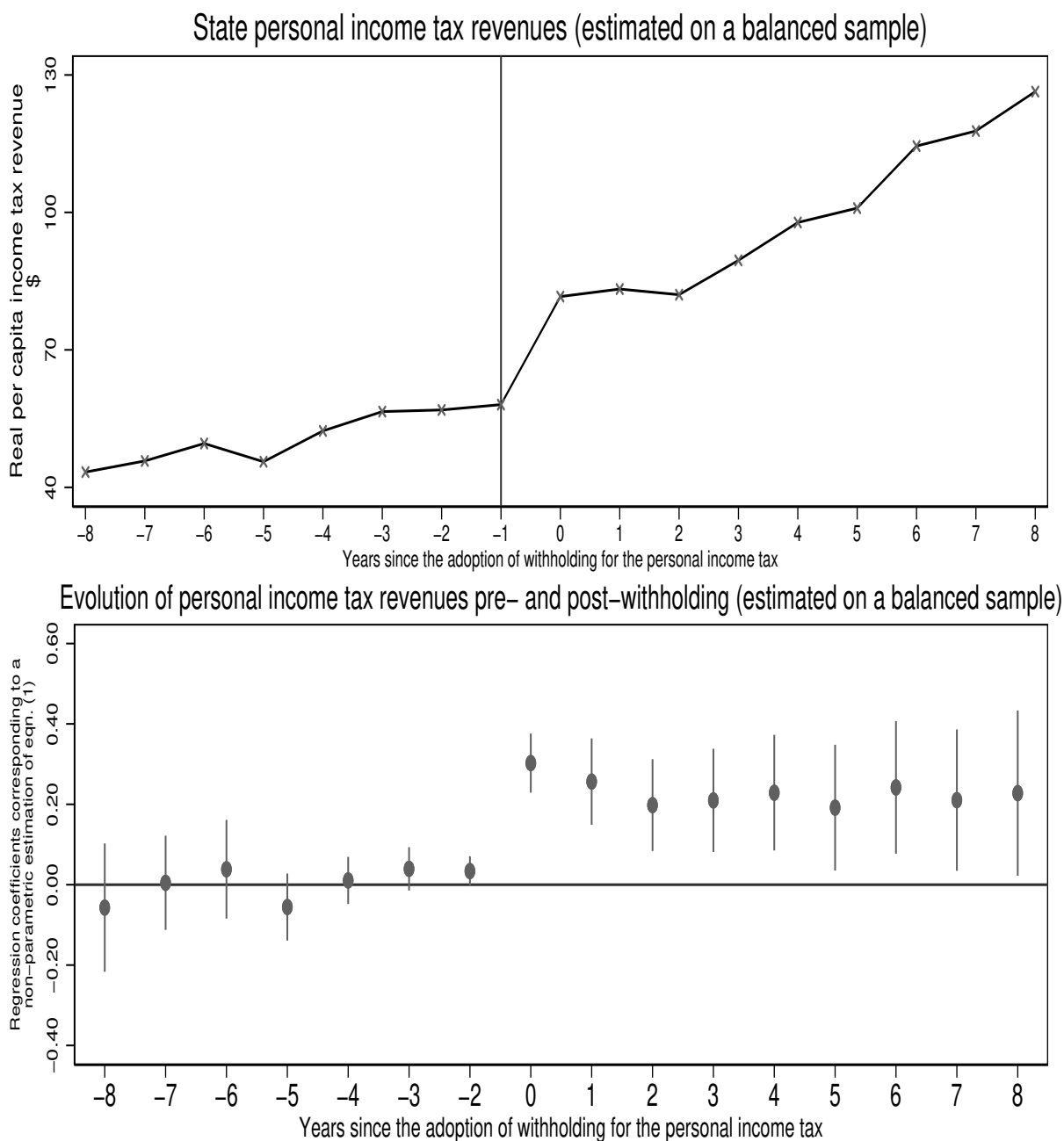
²Recall here that all 29 states in our sample are treated at some point during the sample period and therefore essentially we are exploiting the variation in treatment timing across states.

with the message from Figure 1 which was generated using the conventional two-way fixed effects estimator.

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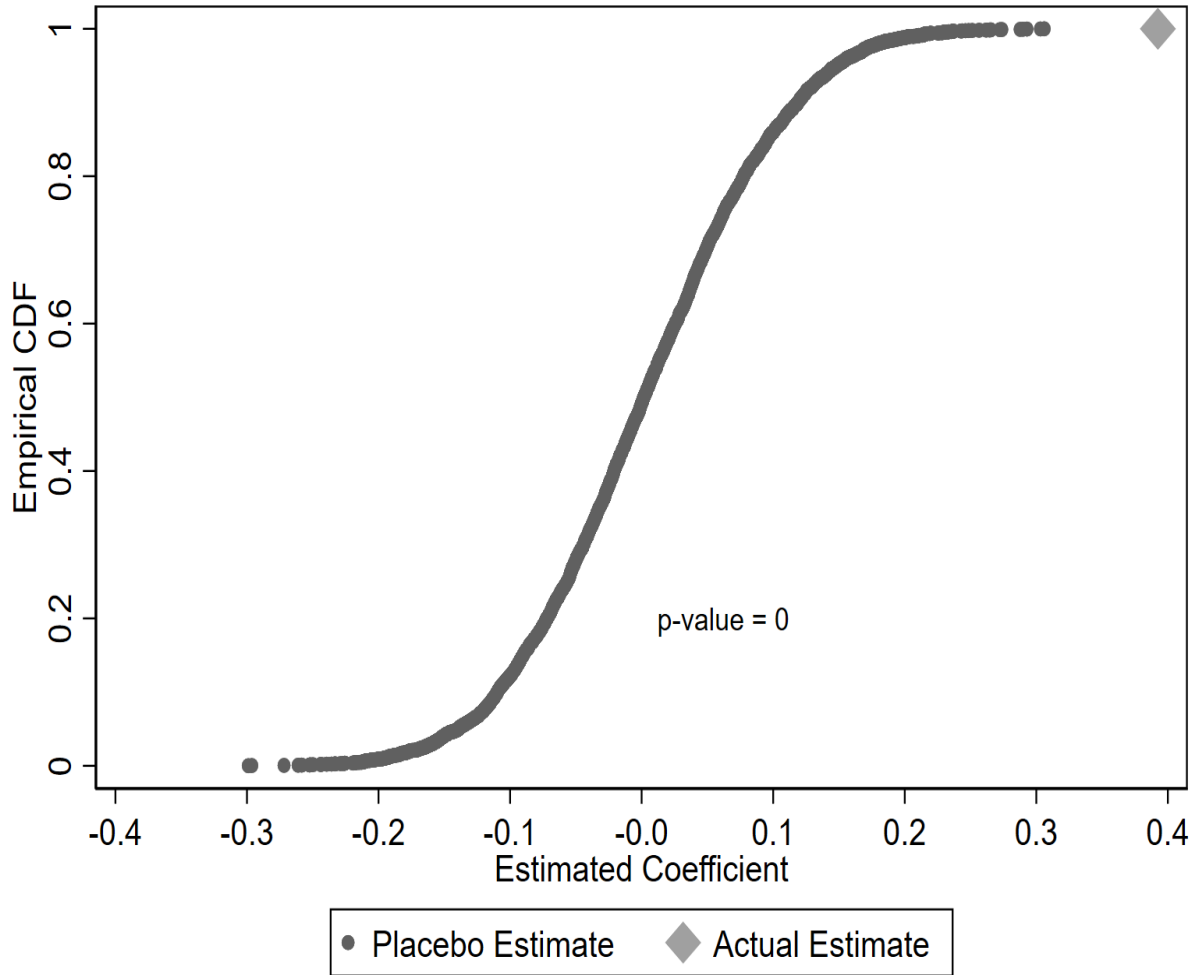
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Figure A.1: Trends in state personal income tax revenues around the time of adoption of withholding



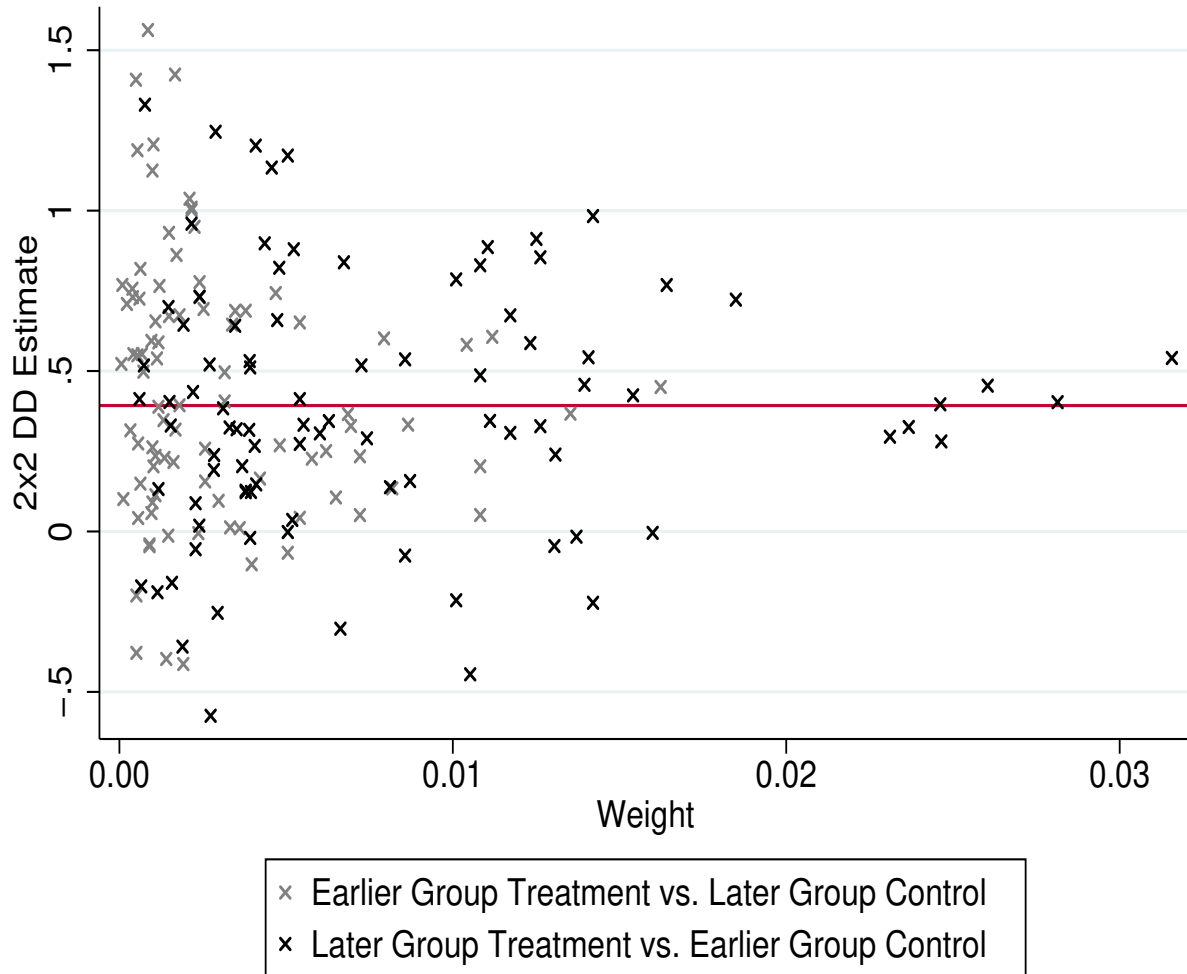
Notes: The top panel plots the average real per capita income tax revenue for those 26 states which adopted withholding after introducing the personal income tax and for which we have at least 9 years of data pre- and post-adoption of withholding. The bottom panel plots the coefficients, along with the 90 percent confidence intervals, from a regression of the log of real per capita income tax revenue on a set of dummy variables for years prior to and after the adoption of withholding, state and year fixed effects, all socio-economic controls, and income tax parameters, with standard errors clustered at the state level. Here as well, the underlying regressions only use observations from the 26 states with at least 9 years of pre- and post-adoption data.

Figure A.2: Permutation test: random assignment of the year of adoption



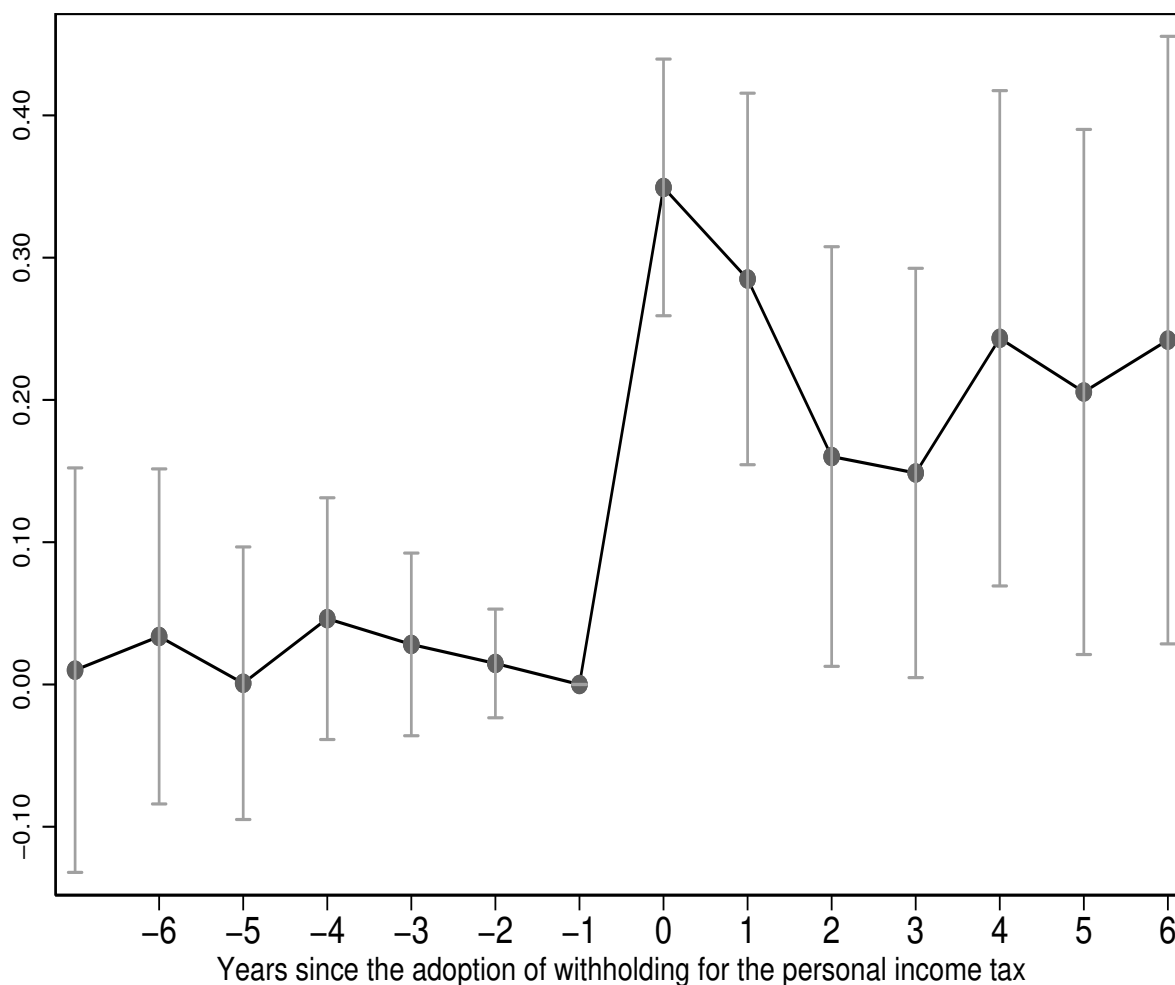
Notes: The placebo adoption years are drawn from a normal distribution, the mean and standard deviation of which are estimated on a support (1948, 1972), i.e. the range of adoption years for all states, except North Dakota. When generating the random placebo years of adoption, the algorithm may pick extreme values indicating the introduction of withholding in say, the year 1920 or 2017. When that happens, we replace such (rare) draws with 1944 (the 1st year in our dataset), and – to keep things symmetric – 1976 (i.e. the last adoption year of $1972 + 4$). We then estimate the specification reported in col. (1) of Table 1 and store the estimated coefficient on the withholding dummy. We run 5000 such permutations and plot the cumulative distribution function above, along with the estimated coefficient from Table 1. We report the p-value associated with the null hypothesis that the estimated effect of withholding is drawn from the same distribution that generates the placebo estimates, whose CDF is plotted above. We obtain similar results if we use an uniform distribution for generating the random placebo years of adoption or instead, continue to use a normal distribution but include 1987 (the year of adoption by North Dakota) to compute the mean and standard deviation of the underlying distribution.

Figure A.3: Bacon decomposition for withholding



Notes: The figure plots each 2x2 DD components from the decomposition theorem developed in Goodman-Bacon (2021) against their weight. The light gray x's are terms in which the earlier adopters act as the treatment group and the later adopters act as the control group. The black x's are terms in which the later adopters act as the treatment group and the earlier adopters act as the control group. The figure notes the average DD estimate and total weight on each type of comparison. The two-way fixed effects estimate, 0.392, corresponding to the most parsimonious specification with state and year fixed effects, equals the average of the y-axis values weighted by their x-axis value. The graph was generated in Stata using the package “bacondecomp” (Goodman-Bacon, Goldring, and Nichols, 2019).

Figure A.4: Event-study plot using an estimator that is robust to heterogeneous treatment effects



Notes: The figure plots the coefficients, along with the 95 percent confidence intervals, from a regression of the log of real per capita income tax revenue on a set of dummy variables and state and year fixed effects, with standard errors clustered at the state level. This is drawn using the Stata package, “did_multiplegt” developed by de Chaisemartin and D’Haultfoeuille (2020) and uses 5,000 bootstraps.

Table A.1: Adoption of the personal income tax, of withholding, and of federal-state information sharing agreements with the IRS

State	Personal income tax	Withholding	Federal-state information sharing
Oregon	1930	1948	1961
Delaware	1917	1949	1965
Vermont	1931	1951	1965
Arizona	1933	1954	1966
Kentucky	1936	1954	1951
Colorado	1937	1954	1952
Idaho	1931	1955	1964
Montana	1933	1955	1951
Maryland	1937	1955	1963
Alabama	1933	1956	1970
Massachusetts	1916	1959	1963
New York	1919	1959	1963
North Carolina	1921	1959	1950
South Carolina	1922	1959	1964
Utah	1931	1959	1961
Georgia	1929	1960	1968
Oklahoma	1915	1961	1963
Missouri	1917	1961	1962
Minnesota	1933	1961	1957
New Mexico	1933	1961	1963
Louisiana	1934	1961	1971
Wisconsin	1911	1962	1950
Virginia	1916	1963	1963
Arkansas	1929	1966	1963
Kansas	1933	1966	1960
Iowa	1934	1966	1962
Mississippi	1912	1968	1966
California	1935	1971	1961
North Dakota	1919	1987	1964

Source: Tables 1, 18, and 34 of Penniman (1980).

Table A.2: Summary statistics (for 29 states with a personal income tax as of 1944; sample period: 1944–2000)

Variable	Obs.	Mean	Standard Dev.	Min.	Max.
Fiscal variables:					
Withholding dummy	1653	0.71	0.45	0	1
Real per capita income tax revenue	1653	177.98	152.33	6.69	805.79
Real per capita corporate income tax revenue	1653	43.38	27.15	0	182.10
Real per capita general sales tax revenue	1653	150.79	111.69	0	494.20
Real per capita selective sales (excise) tax revenue	1653	138.45	36.52	43.29	293.12
Dummy for Audit Exchange Information Agreements	1653	0.69	0.46	0	1
Tax Rates:					
Top income tax rate	1653	7.14	2.76	1	19.8
Lowest income tax rate	1653	1.99	1.17	0.36	6.25
Corporate income tax rate	1653	6.11	2.23	0	12
General sales tax rate	1595	2.80	1.78	0	7
Real motor fuel tax rate (cents per gallon)	1566	9.92	5.51	2	27
Tax base variables:					
Dollar value of standard deduction	1653	1200.78	1234.84	0	7500
Dollar value of personal exemption for single individuals	1653	1015.45	1030.82	0	6000
Extra exemption provided for each taxpayer and/or spouse aged 65 or over	1653	330.69	453.76	0	2229
Number of different brackets	1653	6.25	3.48	0	18
Presence of multiple brackets for different marital/ filing statuses	1653	0.37	0.48	0	1
Percentage of long-term capital gains excluded from AGI	1653	30.87	33.17	0	100
A dummy variable for whether charitable contributions are deductible or not	1653	0.94	0.24	0	1
A dummy variable for whether the federal income tax is deductible or not	1653	0.55	0.5	0	1
Socio-economic characteristics:					
Population (1000s)	1653	3,930	4,678	285	33,988
Real income per capita	1653	9,638	3,582	2,859	21,606
Percent black	1653	11.1	11.8	0.0	47.6
Percent elderly	1653	10.1	2.3	4.5	15.3
Percent urban	1653	62.0	15.7	23.0	94.4
Farm income/ personal income	1653	5.7	6.9	0.0	58.5
Interest & dividend income/ personal income	1653	15.3	4.7	5.0	27.6

Data on when states signed Audit Exchange Information Agreements with the federal government are from Troiano (2018), with sources for the remaining fiscal variables as described in the text in 3.1. The data on state population, personal income, share of farm income, and the share of interest and dividend income in the state's personal income are from the Bureau of Economic Analysis, Regional Accounts Data. The personal income variables were converted into real 1982 dollars by the consumer price index. The farm income series is very volatile and is negative for some states in some years. Because this variable is meant to capture the long-term agricultural vs. industrial character of the state rather than short-term shocks to agricultural incomes, we use three-year moving averages of farm income in the analysis. The percentage of state population that is black, elderly (65+), and lives in urban areas were obtained from Census enumerations. Because Census data are decennial, observation for the years between censuses were generated by simple linear interpolation.

Table A.3: Estimates of the effects of withholding on the personal income tax on balanced panels of varying lengths

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Panel A: Estimated on a sample with at least 5 years of data pre- and post-adoption of withholding				
Withholding dummy	0.358*** (0.0790)	0.316*** (0.0851)	0.278*** (0.0852)	0.266*** (0.0796)
Observations	1596	1596	1596	1596
Number of states included	28	28	28	28
Panel B: Estimated on a sample with at least 7 years of data pre- and post-adoption of withholding				
Withholding dummy	0.332*** (0.0807)	0.301*** (0.0880)	0.262*** (0.0871)	0.253*** (0.0793)
Observations	1539	1539	1539	1539
Number of states included	27	27	27	27
Panel C: Estimated on a sample with at least 9 years of data pre- and post-adoption of withholding				
Withholding dummy	0.294*** (0.0753)	0.259*** (0.0823)	0.223** (0.0822)	0.231*** (0.0794)
Observations	1482	1482	1482	1482
Number of states included	26	26	26	26
Panel D: Estimated on a sample with at least 12 years of data pre- and post-adoption of withholding				
Withholding dummy	0.288*** (0.0787)	0.239** (0.0851)	0.208** (0.0876)	0.231*** (0.0809)
Observations	1311	1311	1311	1311
Number of states included	23	23	23	23
Panel E: Estimated on a sample with at least 13 years of data pre- and post-adoption of withholding				
Withholding dummy	0.244** (0.0941)	0.171 (0.105)	0.133 (0.109)	0.178 (0.104)
Observations	1083	1083	1083	1083
Number of states included	19	19	19	19
Panel F: Estimated on a sample with at least 14 years of data pre- and post-adoption of withholding				
Withholding dummy	0.302** (0.117)	0.249** (0.116)	0.230** (0.106)	0.263** (0.0937)
Observations	1026	1026	1026	1026
Number of states included	18	18	18	18
Panel G: Estimated on a sample with at least 16 years of data pre- and post-adoption of withholding				
Withholding dummy	0.307** (0.126)	0.259* (0.127)	0.230* (0.112)	0.260** (0.0955)
Observations	969	969	969	969
Number of states included	17	17	17	17
Panel H: Estimated on a sample with at least 17 years of data pre- and post-adoption of withholding				
Withholding dummy	0.324* (0.175)	0.281 (0.187)	0.247 (0.157)	0.281* (0.142)
Observations	684	684	684	684
Number of states included	12	12	12	12
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%. For complete notes, refer to Table 1.

Table A.4: Dynamics of the response to the introduction of withholding on personal income tax revenues

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Panel A: Defining short-run as 0–2 years; medium-run as 3–7 years; long-run beyond 7 years				
Short-run effect	0.360***	0.333***	0.304***	0.297***
(between 0-2 years after adoption)	(0.0705)	(0.0650)	(0.0656)	(0.0625)
Medium-run effect	0.391***	0.337***	0.285***	0.276***
(between 3-7 years after adoption)	(0.0941)	(0.0957)	(0.0942)	(0.0903)
Long-run effect	0.550***	0.449***	0.362**	0.354***
(more than 7 years after adoption)	(0.127)	(0.141)	(0.133)	(0.123)
Panel B: Defining short-run as 0–2 years; medium-run as 3–8 years; long-run beyond 8 years				
Short-run effect	0.360***	0.333***	0.304***	0.297***
(between 0-2 years after adoption)	(0.0698)	(0.0646)	(0.0654)	(0.0623)
Medium-run effect	0.407***	0.349***	0.294***	0.284***
(between 3-8 years after adoption)	(0.0975)	(0.0999)	(0.0975)	(0.0932)
Long-run effect	0.559***	0.455***	0.363**	0.359***
(more than 8 years after adoption)	(0.129)	(0.143)	(0.136)	(0.125)
Panel C: Defining short-run as 0–3 years; medium-run as 4–8 years; long-run beyond 9 years				
Short-run effect	0.351***	0.321***	0.288***	0.282***
(between 0-3 years after adoption)	(0.0722)	(0.0682)	(0.0689)	(0.0661)
Medium-run effect	0.434***	0.371***	0.314***	0.302***
(between 4-8 years after adoption)	(0.104)	(0.107)	(0.104)	(0.0986)
Long-run effect	0.570***	0.465***	0.373**	0.369***
(more than 8 years after adoption)	(0.132)	(0.147)	(0.139)	(0.128)
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓
Observations	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29
Time period included in the estimation	1944–2000	1944–2000	1944–2000	1944–2000

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%
State and year fixed effects are included in all columns but the corresponding coefficients have not been shown.
The fiscal and socio-economic control variables included in columns (2)–(4) of both panels A and B are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.

Table A.5: Robustness checks on estimates of the effect of withholding on personal income tax revenues – Allowing for clustering at varying levels

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Panel A: Robust standard errors				
Withholding dummy	0.392*** (0.0777)	0.344*** (0.0774)	0.298*** (0.0769)	0.290*** (0.0732)
Panel B: Clustering standard errors at the state level (Baseline specification)				
Withholding dummy	0.392*** (0.0809)	0.344*** (0.0809)	0.298*** (0.0799)	0.290*** (0.0763)
Panel C: Clustering standard errors at the region level				
Withholding dummy	0.392** (0.0877)	0.344** (0.0730)	0.298** (0.0821)	0.290** (0.0847)
Panel D: Clustering standard errors at the region-year level				
Withholding dummy	0.392** (0.0847)	0.344** (0.0735)	0.298** (0.0822)	0.290** (0.0866)
Observations	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29
Time period included	1944–2000	1944–2000	1944–2000	1944–2000
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%. For complete notes, refer to Table 1.

Table A.6: Robustness checks on estimates of the effect of withholding on personal income tax revenues

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Baseline Specification	0.392*** (0.0809)	0.344*** (0.0809)	0.298*** (0.0799)	0.290*** (0.0763)
RC1: Pre-trend estimates	0.0714 (0.0523)	0.0316 (0.0429)	0.0111 (0.0416)	0.0111 (0.0403)
RC2: Ignoring transitory shocks in years 1 & 2	0.402*** (0.0896)	0.337*** (0.0940)	0.281*** (0.0924)	0.270*** (0.0887)
RC3.1: Including state-specific linear trends	0.293*** (0.0695)	0.288*** (0.0641)	0.230*** (0.0650)	0.221*** (0.0643)
RC3.2: Including state-specific quadratic trends	0.391*** (0.113)	0.367*** (0.107)	0.308*** (0.109)	0.332*** (0.0939)
RC4: Introducing region-by-year FE	0.412*** (0.0741)	0.357*** (0.0795)	0.307*** (0.0780)	0.300*** (0.0767)
RC5: Introducing division-by-year FE	0.411*** (0.0940)	0.337*** (0.107)	0.283** (0.105)	0.275** (0.107)
RC6.1: Weighted regressions with contemporaneous state population as weights	0.389*** (0.0798)	0.341*** (0.0786)	0.298*** (0.0770)	0.293*** (0.0738)
RC6.2: Weighted regressions with state population in 1944 as weights	0.389*** (0.0797)	0.337*** (0.0791)	0.292*** (0.0782)	0.289*** (0.0752)
RC7: Use of random effects (RE) specification	0.394*** (0.0813)	0.363*** (0.0745)	0.312*** (0.0735)	0.301*** (0.0706)
RC8: Distinguishing between states that forgave part of the income tax due				
States that did not forgive tax liabilities	0.429*** (0.0934)	0.359*** (0.0942)	0.300*** (0.0938)	0.280*** (0.0923)
States that forgave part of tax liabilities	0.353*** (0.0582)	0.341** (0.0822)	0.283** (0.0694)	0.245* (0.0785)
State fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * p < 0.10, ** p < 0.05, *** p < 0.01.

Results in this table are variations on the specifications reported in columns (1)–(4) of Table 1. The top row of the current table is the baseline specification. Except where noted, all specifications are estimated using an annual, state-level panel of data for the years 1944–2000. A list of the states included in each Census region and division is available at https://www2.census.gov/geo/docs/maps-data/maps/reg_div.txt. Region-by-year FE in RC4 and division-by-year FE in RC5 are for the four Census regions and nine Census divisions respectively. The states which forgave a portion of their tax liabilities are California (20 percent of 1971 liabilities), New York (All 1958 liabilities except those on trusts, estates, and capital gains), Minnesota (Roughly 75 percent of 1961 liabilities), and Wisconsin (65 percent of 1961 liabilities).

Table A.7: Estimates of the effects of withholding, controlling for potential changes to aspects of the personal income tax base

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependent variable: Log of real per capita personal income tax revenues									
Withholding dummy	0.292*** (0.0714)	0.290*** (0.0762)	0.289*** (0.0753)	0.289*** (0.0763)	0.287*** (0.0763)	0.292*** (0.0766)	0.278*** (0.0732)	0.289*** (0.0761)	0.278*** (0.0663)
Dollar value of standard deduction	0.00545 (0.00411)								0.00435 (0.00490)
Dollar value of personal exemption for single individuals		0.000328 (0.00607)							-0.00537 (0.00645)
Extra exemption provided for each taxpayer and/or spouse 65 or over			0.00136 (0.00358)						0.00310 (0.00376)
Number of different brackets				0.0184 (0.0299)					-0.0355 (0.0335)
Presence of multiple brackets for different marital/ filing statuses					-0.00730 (0.0138)				0.00410 (0.0136)
Percentage of long-term capital gains excluded from AGI						-0.0103** (0.00408)			-0.0130*** (0.00404)
A dummy variable for whether charitable contributions are deductible or not							0.0585*** (0.0158)		0.0861*** (0.0250)
A dummy variable for whether the federal income tax is deductible or not							-0.00936 (0.0119)		-0.0133 (0.0129)
Observations	1653	1653	1653	1653	1653	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29	29	29	29	29	29
Time period included	1944–2000	1944–2000	1944–2000	1944–2000	1944–2000	1944–2000	1944–2000	1944–2000	1944–2000
State and year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Socio-economic controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Top income tax rate	✓	✓	✓	✓	✓	✓	✓	✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets	✓	✓	✓	✓	✓	✓	✓	✓	✓

Std. errors clustered by state in parentheses * significant at 10%, ** significant at 5%; *** significant at 1%
State and year fixed effects, socio-economic controls, the top and bottom personal tax rate, top and bottom personal tax bracket, and interactions of the corresponding tax rate with tax bracket are included in all columns but the corresponding coefficients have not been shown. The fiscal and socio-economic control variables included are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs. The base variables have also been introduced in log form. As the base variables frequently bunch at zero, we add 0.01 prior to taking logs. Our results are robust to alternative transformations.

Table A.8: Estimates of the effects of withholding on aspects of the personal income tax base

	(1)	(2)	(3)
Panel A: Dependent variable: Dollar value of standard deduction			
Withholding dummy	-0.474 (1.460)	-0.232 (1.369)	-0.586 (0.962)
Panel B: Dependent variable: Dollar value of personal exemption for single individuals			
Withholding dummy	0.200 (1.067)	0.460 (1.108)	0.0145 (0.661)
Panel C: Dep. variable: Extra exemption provided for each taxpayer and/or spouse aged 65 or over			
Withholding dummy	0.886 (1.492)	0.713 (1.527)	1.013 (1.093)
Panel D: Dependent variable: Number of different brackets			
Withholding dummy	0.174 (0.118)	0.143 (0.0935)	0.179* (0.0912)
Panel E: Dependent variable: Presence of multiple brackets for different marital/ filing statuses			
Withholding dummy	-0.289 (0.265)	-0.318 (0.203)	-0.341** (0.160)
Panel F: Dependent variable: Percentage of long-term capital gains excluded from AGI			
Withholding dummy	0.414 (0.839)	0.367 (0.796)	0.679 (0.639)
Panel G: Dep. variable: A dummy variable for whether charitable contributions are deductible or not			
Withholding dummy	0.201 (0.244)	0.220 (0.204)	0.114 (0.109)
Panel H: Dependent variable: A dummy variable for whether the federal income tax is deductible or not			
Withholding dummy	0.0222 (0.278)	-0.127 (0.265)	-0.0474 (0.204)
Observations (for all eight panels A through H)	1653	1653	1653
Number of states included in the estimation	29	29	29
Time period included	1944–2000	1944–2000	1944–2000
State and year fixed effects	✓	✓	✓
Socio-economic controls		✓	✓
State-specific linear time trends			✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%. Data on tax base variables (e.g., personal exemption, standard deduction, etc.), are based on Bakija (2017). State and year fixed effects are included in all columns and state-specific linear time trends in col. (3) but the corresponding coefficients have not been shown. The fiscal and socio-economic control variables included are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs. The base variables have also been introduced in log form. As the base variables frequently bunch at zero, we add 0.01 prior to taking logs. Our results are robust to alternative transformations.

Table A.9: Estimates of the effects of withholding on the personal income tax, controlling for the passage of the Industrial Development Bonds program discussed in Jensen (2019)

	(1)	(2)	(3)	(4)
Panel A: States that adopted the IDB program				
Dependent variable: Log of real per capita personal income tax revenues				
Withholding dummy	0.340*** (0.0933)	0.276*** (0.0971)	0.235** (0.0972)	0.227** (0.0966)
Observations	1254	1254	1254	1254
Number of states included in the estimation	22	22	22	22
Panel B: States that did not adopt the IDB program				
Dependent variable: Log of real per capita personal income tax revenues				
Withholding dummy	0.489*** (0.0952)	0.414*** (0.0620)	0.353*** (0.0596)	0.368*** (0.0594)
Observations	399	399	399	399
Number of states included in the estimation	7	7	7	7
p-value from a t-test that the effects of withholding are equal across the sub-samples in Panels A and B:	0.2238	0.2022	0.2687	0.1868
Panel C: States that adopted the IDB program; examining the differential impact of the IDB program				
Dependent variable: Log of real per capita personal income tax revenues				
Withholding dummy	0.347*** (0.0875)	0.330*** (0.0859)	0.276*** (0.0797)	0.280*** (0.0847)
Dummy that the IDB program was upheld by the state supreme court	-0.0556 (0.0962)	0.0817 (0.108)	0.0330 (0.119)	0.0738 (0.111)
Withholding dummy X Dummy that the IDB program was upheld by the state supreme court	-0.0294 (0.101)	-0.124 (0.111)	-0.0963 (0.121)	-0.123 (0.101)
Observations	1254	1254	1254	1254
Number of states included in the estimation	22	22	22	22
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%
State and year fixed effects are included in all columns but the corresponding coefficients have not been shown.
The fiscal and socio-economic control variables included in cols. (2)–(4) are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.
The year in which the IDB program was upheld by each state supreme court is sourced from Jensen (2019) (Ref: Table B.1: Industrial Development Bonds Program Legal Timing, pp. 34 and 35).

Table A.10: Estimates of the effects of withholding on the personal income tax, accounting for heterogeneity in when withholding was introduced and the level of experience states had with the personal income tax (PIT)

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
First sample-split: Based on the year of adoption of withholding, with states adopting withholding prior to 1959 classified as early adopters and those adopting after 1959 classified as late adopters				
Panel A1: Effects of withholding in states that were “early” adopters of withholding				
Withholding dummy	0.351*** (0.0927)	0.341*** (0.0720)	0.309*** (0.0738)	0.306*** (0.0557)
Observations	627	627	627	627
Number of states included in the estimation	11	11	11	11
Panel A2: Effects of withholding in states that were “late” adopters of withholding				
Withholding dummy	0.245** (0.0992)	0.174 (0.112)	0.128 (0.114)	0.170 (0.108)
Observations	1026	1026	1026	1026
Number of states included in the estimation	18	18	18	18
p-value from a t-test that the effects of withholding are equal across the sub-samples in Panels A1 and A2:	0.4083	0.1854	0.1588	0.2395
Second sample-split: Based on the level of experience with the state PIT, where experience is quantified as the number of years elapsed between the adoption of the state PIT and the adoption of withholding				
Panel B1: Effects of withholding in states that had more experience with the state PIT				
Withholding dummy	0.361** (0.123)	0.233** (0.102)	0.202* (0.108)	0.252** (0.0937)
Observations	855	855	855	855
Number of states included in the estimation	15	15	15	15
Panel B2: Effects of withholding in states that had less experience with the state PIT				
Withholding dummy	0.442*** (0.0869)	0.413*** (0.0610)	0.385*** (0.0616)	0.332*** (0.0582)
Observations	798	798	798	798
Number of states included in the estimation	14	14	14	14
p-value from a t-test that the effects of withholding are equal across the sub-samples in Panels B1 and B2:	0.5700	0.1074	0.1184	0.4440
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%
State and year fixed effects are included in all columns but the corresponding coefficients have not been shown.
The fiscal and socio-economic control variables included in cols. (2)–(4) are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.

Table A.11: Estimates of the effects of withholding on federal personal income tax collections and number of tax returns filed

	(1)	(2)	(3)
Panel A: Dependent variable: Log of real federal personal per capita income tax revenue			
Withholding dummy	0.0950* (0.0533)	0.0772*** (0.0272)	0.0391 (0.0319)
Panel B: Dependent variable: Log of real federal personal per capita payroll tax revenue			
Withholding dummy	0.0898* (0.0486)	0.0689** (0.0250)	0.0172 (0.0304)
Panel C: Dependent variable: Log of real federal personal per capita income and payroll tax revenue			
Withholding dummy	0.103* (0.0524)	0.0737*** (0.0263)	0.0378 (0.0318)
Panel D: Dependent variable: Log of number of federal income tax returns filed annually			
Withholding dummy	-0.0331 (0.0276)	-0.0106 (0.0109)	-0.0139* (0.00680)
Panel E: Dependent variable: Log of number of federal income tax units			
Withholding dummy	-0.0597** (0.0258)	-0.0122** (0.00582)	-0.00321 (0.00482)
Observations (for all five panels A through E)	1653	1653	1653
Number of states included in the estimation	29	29	29
Time period included	1944–2000	1944–2000	1944–2000
State and year fixed effects	✓	✓	✓
Socio-economic controls		✓	✓
State-specific linear time trends			✓

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%. Data on federal tax collections by state are sourced from IRS Data Books. Data on the number of federal tax returns filed annually and number of federal income tax units are sourced from U.S. State-Level Income Inequality Data compiled by Mark W. Frank and available at: https://www.shsu.edu/eco_mwf/inequality.html. State and year fixed effects are included in all columns and state-specific linear time trends in col. (3) but the corresponding coefficients have not been shown. The fiscal and socio-economic variables included are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.

Table A.12: Estimates of the relationship between change in the number of returns filed annually and average AGI per return – estimated using county-level data for California

	(1)	(2)	(3)
Dependent variable: Year-on-year change in the number of state income tax returns filed annually in California			
Panel A: Average AGI per return introduced in levels and separate regressions run for the pre- and post-withholding periods			
	Estimated using pre-withholding years	Estimated using post-withholding years	p-value from a t-test that the coefficients in cols. (1) and (2) are equal to each other
Average AGI per return (measured as of 1968) (in levels)	0.000425	-0.00119**	0.0643
Observations	(0.000534)	(0.000510)	
	132	88	
Panel B: Average AGI per return introduced in logs and separate regressions run for the pre- and post-withholding periods			
	Estimated using pre-withholding years	Estimated using post-withholding years	p-value from a t-test that the coefficients in cols. (1) and (2) are equal to each other
Average AGI per return (measured as of 1968) (in logs)	3.116	-11.40**	0.0864
Observations	(5.424)	(4.585)	
	132	88	
Panel C: Average AGI per return introduced in levels (1) and logs (2) and a single regression run for the entire sample period			
Withholding dummy	27.13***	144.6*	
	(9.683)	(79.19)	
Average AGI per return (measured as of 1968) (in levels)	0.000425		
	(0.000536)		
Average AGI per return (measured as of 1968) (in levels) X Withholding dummy	-0.00162*		
	(0.000881)		
Average AGI per return (measured as of 1968) (in logs)		3.116	
		(5.440)	
Average AGI per return (measured as of 1968) (in logs) X Withholding dummy		-14.52*	
Observations	220	(8.526)	
		220	

Std. errors clustered by county in parentheses * p < 0.10, ** p < 0.05, *** p < 0.01.

Data on the number of state income tax returns filed by county in California are available for the period 1968–1973 along with the aggregate AGI reported on those returns. The average AGI per return as of 1968 for each county is calculated as the aggregate AGI reported on all 1968 returns from that county divided by the number of returns filed for that county. Given that California adopted withholding in 1971 and it came into effect on 1st January 1972, the year-on-year change in number of returns between 1968 and 1969, 1969 and 1970, 1970 and 1971 are our pre-withholding changes, whereas changes between years 1971 and 1972 and 1972 and 1973 are our post-withholding changes. The regressions are estimated using only those 44 counties for which we are able to construct an integrated time series for 1968–1973. For the remaining California counties (which are smaller in population), disaggregated data are not available for the pre-withholding years and accordingly, they are not used in the estimation. We are grateful to the California FTB for making these data available to us.

Table A.13: Dynamics of the response to the introduction of withholding using an estimator that is robust to heterogeneous treatment effects

	(1)	(2)	(3)	(4)
Dependent variable: Log of real per capita personal income tax revenues				
Panel A: Using 4 leads and 4 lags				
Average effect of withholding	0.237*** (0.0610)	0.235*** (0.0606)	0.235*** (0.0611)	0.229*** (0.0635)
p-value from joint test that all placebos requested are equal to 0	0.723	0.626	0.688	0.718
Panel B: Using 5 leads and 5 lags				
Average effect of withholding	0.232*** (0.0663)	0.228*** (0.0667)	0.228*** (0.0676)	0.220*** (0.0708)
p-value from joint test that all placebos requested are equal to 0	0.739	0.348	0.456	0.496
Panel C: Using 6 leads and 6 lags				
Average effect of withholding	0.233*** (0.0722)	0.225*** (0.0740)	0.226*** (0.0753)	0.216*** (0.0786)
p-value from joint test that all placebos requested are equal to 0	0.766	0.470	0.585	0.625
Panel D: Using 7 leads and 7 lags				
Average effect of withholding	0.226*** (0.0765)	0.214*** (0.0794)	0.215*** (0.0812)	0.205** (0.0850)
p-value from joint test that all placebos requested are equal to 0	0.375	0.434	0.502	0.526
Panel E: Using 8 leads and 8 lags				
Average effect of withholding	0.218*** (0.0841)	0.203** (0.0875)	0.203** (0.0895)	0.192** (0.0943)
p-value from joint test that all placebos requested are equal to 0	0.396	0.534	0.608	0.633
State and year fixed effects	✓	✓	✓	✓
Socio-economic controls		✓	✓	✓
Top income tax rate			✓	✓
Top and bottom income tax rates, brackets, and the interactions of rates with brackets				✓
Observations	1653	1653	1653	1653
Number of states included in the estimation	29	29	29	29
Time period included in the estimation	1944–2000	1944–2000	1944–2000	1944–2000

The results were generated using the estimator, `did_multiplegt`, discussed in de Chaisemartin and D'Haultfoeuille (2020). Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%

State and year fixed effects are included in all columns but the corresponding coefficients have not been shown. The fiscal and socio-economic control variables included in columns (2)–(4) of both panels A and B are real income per capita, share of farm income, share of capital income, population, percent elderly, percent Black, and percent urban and are in logs.